Combinatorics of the critical loci of deep linear networks

March 12, 2026

Abstract

We study the critical loci of deep linear networks i.e. the set of parameters at which the gradient of the loss vanishes. The set of vanishing gradients can be described as an algebraic variety described by cyclic vanishing conditions. This allows us to study the fiber of the gradient map using cyclic quiver representation theory and stratify the critical loci into orbits parametrized by rank patterns i.e. the ranks of partial products of the deep linear networks weights. These orbits allow to better understand the geometry of the critical loci as the codimension of the largest irreducible component relates to the volume scaling around the critical loci of the deep linear network.

1 Introduction

1.1 Related works

1.2 Contributions

2 Setup and notations

Definition 2.1. A L layers Deep Linear Network (DLN) is a function:

$$f : \mathbb{R}^{d_0} \times \prod_{l=0}^{L-1} \mathbb{R}^{d_l \times d_{l+1}} \to \mathbb{R}^{d_L}$$
$$(x, W_1, ..., W_L) \mapsto W_L ... W_1 x$$

The dimension d_0 is the dimension of the input data and the dimensions $d_l,\ 1 \leq l \leq L$ are the width of the layers. A Deep Linear Network is a multi-linear function which is linear in each of the weight matrix W_l and the data.

Definition 2.2. Consider the product map:

$$\mu : \prod_{l=0}^{L-1} \mathbb{R}^{d_l \times d_{l+1}} \to \mathbb{R}^{d_0 \times d_L}$$
$$W := (W_1, ..., W_L) \mapsto W_L ... W_1$$

This function μ maps the parameter space $\Omega := \{(W_1, ..., W_L) \in \prod_{l=0}^{L-1} \mathbb{R}^{d_l \times d_{l+1}}\}$ to the function space $\mathcal{F} \cong \mathbb{R}^{d_0 \times d_L}$ of a Deep Neural Network. A DLN is a matrix-monomial (so non-linear) in the parameter space Ω (the space above) and linear map in the function space $\mathcal{F}$ (the space below).

Definition 2.3. Define the variety of matrices of rank r:

$$\Sigma_d^r := \{W \in \Omega | rk(W_L...W_1) = r\}$$

Consider the following group action of $GL_d = \Pi_{l=L}^0 GL_{d_l} = GL_{d_L} \times GL_h \times GL_{d_0}$:

$$(W_1, \dots, W_L) \mapsto (g_1 W_1 g_0^{-1},\ g_2 W_2 g_1^{-1},\ \dots,\ g_L W_L g_{L-1}^{-1})$$

We can also reparametrize Σ_d^r by the following:

$$\Sigma_d^r = \{W \in \Omega, Z_l \in \mathbb{R}^{d_l - r \times d_{l+1} - r} | W_l \sim \begin{pmatrix} I_r & 0 \\ 0. & Z_l \end{pmatrix}, Z_L...Z_1 = 0\}$$

Note that the latter product over Z's–the *residuals*–can be defined by the algebraic variety:

$$\Sigma_0^{d-r} = \{Z_l \in \mathbb{R}^{d_l - r \times d_{l+1} - r} | Z_L...Z_1 = 0\}$$

Definition 2.4. Define the algebraic variety Γ by the following cyclic conditions:

$$\Gamma_d^r = \{Z_l \in \mathbb{R}^{d_l - r \times d_{l+1} - r} \mid Z_{l-1} \cdots Z_1 \, \Sigma_{XY} U_Q \, Z_L \cdots Z_{l+1} = 0\}$$

Define the algebraic variety

$$\Gamma_{d,XY} := \Sigma_0^{d-r} \cap \Gamma_d^r$$

also defined by a set of vanishing polynomial cyclic conditions.

Fix a depth $L \geq 2$, widths $(d_0, \dots, d_L)$, and an integer $0 \leq r \leq \min_\ell d_\ell$. Let

$$\Omega \ := \ \prod_{\ell=1}^{L} \mathrm{Mat}_{d_\ell \times d_{\ell-1}}(\mathbb{R})$$

and let the base–change group

$$G \ := \ \prod_{\ell=0}^{L} GL(d_\ell, \mathbb{R})$$

act on Ω by

$$(g \cdot W)_\ell \ := \ g_\ell \, W_\ell \, g_{\ell-1}^{-1}. \tag{1}$$

We also fix the data-dependent matrices appearing in the normal form of critical points (e.g. from the DLN second-order analysis):

$$U_S \in \mathbb{R}^{d_L \times r}, \qquad U_Q \in \mathbb{R}^{d_L \times (d_L - r)}, \qquad \sigma \ := \ \Sigma_{XY} U_Q \in \mathrm{Hom}(\mathbb{R}^{d_L - r}, \mathbb{R}^{d_0 - r}).$$

(Here σ is the distinguished "closing arrow" in the cyclic-quiver model.)

Proposition 2.5. From the paper [?], we can write the set of rank r critical points of a DLN as:

$$\mathcal{C}_r = \{W \in \Omega, Z_l \in \mathbb{R}^{d_l - r \times d_{l+1} - r} | W_L = [U_S, U_Q Z_L], W_l \sim \begin{pmatrix} I_r & 0 \\ 0. & Z_l \end{pmatrix}, W_1 = [\Sigma_X^{-1} \Sigma_{XY} U_S, Z_1^\top]^\top, Z_l \in \Gamma_{d,XY}\}$$

Which suggest that studying the geometry of the critical loci amounts to studying the geometry of $\Gamma_{d,XY}$.

3 Critical loci as a bundle over Γ

A gauge-fixed slice of rank-r critical points. Define the map $\iota : \mathcal{Z} \to \Omega$ by putting residual blocks on the diagonal:

$$\iota(Z)_\ell \;:=\; \begin{pmatrix} I_r & 0 \\ 0 & Z_\ell \end{pmatrix} \in \mathrm{Mat}_{d_\ell \times d_{\ell-1}}(\mathbb{R}), \qquad \ell = 2, \dots, L-1, \tag{2}$$

and set the endpoints to match the standard DLN critical-point normal form:

$$\iota(Z)_L \;:=\; [\,U_S,\; U_Q Z_L\,] \in \mathrm{Mat}_{d_L \times d_{L-1}}(\mathbb{R}), \qquad \iota(Z)_1 \;:=\; \begin{pmatrix} U_S^\top \Sigma_{YX} \Sigma_X^{-1} \\ Z_1 \end{pmatrix} \in \mathrm{Mat}_{d_1 \times d_0}(\mathbb{R}). \tag{3}$$

(Only the block structure matters for what follows; any equivalent fixed endpoint embedding works.) Define the *gauge-fixed critical slice*

$$\mathcal{S}_{r,\sigma} \;:=\; \iota(\Gamma_{d,XY}) \;\subset\; \Omega. \tag{4}$$

Hidden-layer gauge group and its slice stabilizer. Let

$$G_h \;:=\; \prod_{\ell=1}^{L-1} GL(d_\ell, \mathbb{R})$$

act on Ω via (1) with $g_0 = g_L = I$ (endpoints fixed). Define the subgroup $H_{r,\sigma} \le G_h$ consisting of those hidden reparametrizations that preserve the block decomposition $\mathbb{R}^{d_\ell} \cong \mathbb{R}^r \oplus \mathbb{R}^{d_\ell - r}$ for every hidden layer *and* preserve the endpoint embeddings (3):

$$H_{r,\sigma} \;:=\; \left\{ h = (h_1, \dots, h_{L-1}) \in G_h \;\middle|\; h_\ell = \begin{pmatrix} A & 0 \\ 0 & K_\ell \end{pmatrix},\; A \in GL(r,\mathbb{R}),\; K_\ell \in GL(d_\ell - r, \mathbb{R}),\; \text{and } h \cdot \mathcal{S}_{r,\sigma} = \mathcal{S}_{r,\sigma} \right\}. \tag{5}$$

(Equivalently: $H_{r,\sigma}$ is the setwise stabilizer of $\mathcal{S}_{r,\sigma}$ inside G_h). The induced action of $H_{r,\sigma}$ on $\mathcal{Z}$ is the residual base-change action: for $h_\ell = \left(\begin{smallmatrix} A & 0 \\ 0 & K_\ell \end{smallmatrix}\right)$ we set

$$(h \cdot Z)_\ell \;:=\; K_\ell \, Z_\ell \, K_{\ell-1}^{-1} \qquad (\ell = 2, \dots, L-1), \tag{6}$$

with the obvious endpoint conventions for $\ell = 1$ and $\ell = L$ determined by the fixed endpoint embeddings (3). By construction, $\Gamma_{d,XY} \subset \mathcal{Z}$ is $H_{r,\sigma}$-stable (this is exactly the meaning of $h \cdot \mathcal{S}_{r,\sigma} = \mathcal{S}_{r,\sigma}$).

Theorem 3.1. Let $C_r \subset \Omega$ denote the set of rank-r first-order critical points in the normal form of the DLN second-order characterisation, i.e. the set described by the existence of residual blocks $Z = (Z_1, \dots, Z_L)$ such that

$$W_L = [U_S,\, U_Q Z_L], \qquad W_\ell \sim \begin{pmatrix} I_r & 0 \\ 0 & Z_\ell \end{pmatrix} \; (\ell = 2, \dots, L-1), \qquad W_1 = \begin{pmatrix} U_S^\top \Sigma_{YX} \Sigma_X^{-1} \\ Z_1 \end{pmatrix}, \qquad Z \in \Gamma_{d,XY}$$

Then the following hold.

(1) Orbit-saturation description.

$$C_r = G_h \cdot S_{r,\sigma}. \tag{22}$$

(2) Associated bundle model (set-level). Define $\Phi\colon G_h \times \Gamma_{d,XY} \to \Omega$ by

$$\Phi(g, Z) := g \cdot \iota(Z), \tag{23}$$

Then $\Phi(G_h \times \Gamma_{d,XY}) = C_r$ and Φ is invariant under the right action of $H_{r,\sigma}$ on $G_h \times \Gamma_{d,XY}$ given by

$$(g, Z) \cdot h := (gh,\ h^{-1} \cdot Z). \tag{24}$$

Hence Φ descends to a surjective map

$$\bar{\Phi}\colon G_h \times_{H_{r,\sigma}} \Gamma_{d,XY} \longrightarrow C_r.$$

Define the *transporter into the slice* at a point $Z' \in \Gamma_{d,XY}$ by

$$\mathrm{Trans}_{G_h}\big(\iota(Z'),\, S_{r,\sigma}\big) := \big\{g \in G_h : g \cdot \iota(Z') \in S_{r,\sigma}\big\}.$$

Define the *principal slice locus*

$$\Gamma^{\circ}_{d,XY} := \big\{Z' \in \Gamma_{d,XY} \;\big|\; \mathrm{Trans}_{G_h}\big(\iota(Z'),\, S_{r,\sigma}\big) = H_{r,\sigma}\big\},$$

and let $C_r^{\circ} := \Phi(G_h \times \Gamma^{\circ}_{d,XY})$. Then the restriction

$$\bar{\Phi}\colon G_h \times_{H_{r,\sigma}} \Gamma^{\circ}_{d,XY} \xrightarrow{\sim} C_r^{\circ}$$

is an isomorphism. In particular, on the principal slice locus one the identification:

$$C_r^{\circ} \;\cong\; G_h \times_{H_{r,\sigma}} \Gamma^{\circ}_{d,XY} \qquad \text{(on the principal slice locus).} \tag{25}$$

(3) Quotient of critical points by hidden gauge. On the same principal slice locus,

$$C_r^{\circ}/G_h \;\cong\; \Gamma^{\circ}_{d,XY}/H_{r,\sigma}. \tag{26}$$

The proof of the latter proposition can be found appendix 5.3.

4 Cyclic quiver representations

4.1 Critical loci as cyclic quivers

A cyclic quiver is a quiver with at least one oriented cycle. In this paper, we will be interested in a quiver which is exactly one cycle here noted C_n.

Definition 4.1 (Cyclic quiver)**.** A *cyclic quiver* (or *oriented cycle*) on $n \geq 1$ vertices is the quiver C_n with vertex set

$$P = \{0, 1, \ldots, n-1\}$$

and arrow set

$$E = \{a_i : i-1 \to i \mid i = 1, \ldots, n-1\} \cup \{a_0 : n-1 \to 0\},$$

i.e. the vertices are arranged in a directed cycle.

Definition 4.2 (Representation of a cyclic quiver)**.** A *representation* of the cyclic quiver C_n (arrows of C_n are of the opposite orientation) over a field k is the data of vector spaces

$$(V_0, \dots, V_{n-1})$$

together with linear maps

$$\phi_i : V_{i-1} \to V_i \quad (i = 1, \dots, n-1), \qquad \phi_0 : V_{n-1} \to V_0,$$

i.e. a representation of the opposite quiver C_n.

Definition 4.3 (Cyclic product maps)**.** Fix $L \geq 1$ and consider the cyclic quiver C_{L+1} on vertices

$$\{0, 1, \dots, L\}.$$

Let

$$V_l \cong \mathbb{R}^{d_l - r}, \qquad l = 0, \dots, L,$$

and consider a representation of the quiver C_{L+1} given by linear maps

$$\phi_l : V_{l-1} \to V_l, \qquad l = 0, \dots, L,$$

where indices are taken modulo $L+1$. For $\phi = (\phi_0, \dots, \phi_L) \in \mathrm{Rep}_d(C_{L+1})$ and $l = 0, \dots, L$, define

$$d\mu_l(\phi) := \phi_{l-1} \cdots \phi_1 \, \phi_0 \, \phi_L \cdots \phi_{l+1} \in \mathrm{Hom}(V_l, V_{l-1}) \mod L+1.$$

Each map $d\mu_l(\phi)$ is obtained by composing the arrows along the unique oriented path of length L in the cycle C_{L+1} that omits the arrow ϕ_l. Equivalently, the family $(d\mu_l)_{l=0}^{L}$ enumerates all directed paths of length L in the cycle with $L+1$ vertices.

Definition 4.4 (Cyclic vanishing variety)**.** With the notation of Definition 4.3, define the closed subvariety

$$\Gamma_{d,L+1} := \Big\{ \phi \in \mathrm{Rep}_d(C_{L+1}) \ \Big| \ d\mu_l(\phi) = 0 \quad \text{for all } l = 0, \dots, L \Big\}.$$

Thus $\Gamma_{d,L+1}$ consists of cyclic quiver representations for which all length-L compositions around the cycle vanish.

Remark 4.5 (The cyclic relations as the zero fibre of a functor)**.** The family of maps $d\mu_l$ defined above can be assembled into a single morphism between representation spaces. Namely, define

$$F : \mathrm{Rep}_d(C_{L+1}) \longrightarrow \mathrm{Rep}_d(C_{L+1}^{op})$$

by

$$F(\phi) := \big(d\mu_0(\phi), \dots, d\mu_L(\phi) \big).$$

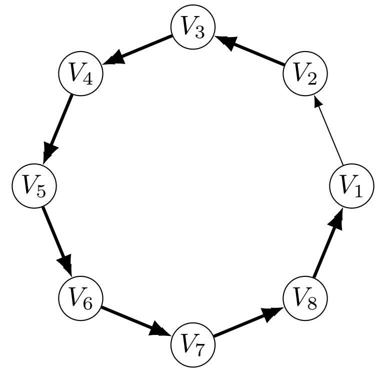
Each component $d\mu_l(\phi)$ is obtained by composing the maps ϕ_i along the unique oriented path of length L in the cycle which omits the arrow ϕ_l. Consequently

$$d\mu_l(\phi) \in \mathrm{Hom}(V_l, V_{l-1})$$

defines an arrow in the opposite quiver C_{L+1}^{op}. With this notation, the variety $\Gamma_{d,L+1}$ can be interpreted as the zero fibre of F:

$$\Gamma_{d,L+1} = F^{-1}(0).$$

Thus $\Gamma_{d,L+1}$ consists precisely of those cyclic representations for which all length-L compositions around the cycle vanish.



$d\mu_1 : V_2 \to V_1$ (composition along the cycle, omitting ϕ_2)

Figure 1: The cyclic quiver C_n (illustrated here with $n = 8$). The thick arrows indicate the path whose composition gives a typical $d\mu_l$ map along the cycle (example: $d\mu_2 : V_2 \to V_1$).

Definition 4.6 (Specialization by the closing arrow)**.** Identifying the residual parameters of a deep linear network with the quiver maps

$$\phi_l = Z_l, \qquad l = 0, \dots, L,$$

we now specialize the closing arrow by fixing

$$\phi_0 = \sigma := \Sigma_{XY} U_Q : V_L \to V_0.$$

The algebraic variety relevant for the critical loci of the DLN is then

$$\Gamma_{d,\sigma} = \Gamma_{d,L+1}\big|_{\phi_0=\sigma}.$$

And

$$\Gamma_{d,XY} = \Gamma_{d,\sigma}\big|_{\phi_l=Z_l \text{ for } l=1,\dots,L}.$$

Equivalently, $\Gamma_{d,XY}$ is the space of cyclic quiver representations satisfying the vanishing relations $d\mu_l = 0$ with the distinguished arrow ϕ_0 fixed to σ and the ϕ_l for $l = 1, \dots, L$ identified with the residual parameters Z_l of the DLN.

Proposition 4.7 (Equivariance of the cyclic composition maps)**.** Let C_{L+1} be the cyclic quiver on vertices $\{0, \dots, L\}$, let $\mathrm{Rep}_d(C_{L+1})$ be its representation space with $V_l \cong \mathbb{R}^{d_l - r}$. Define the base–change group

$$G_d := \prod_{l=0}^{L} \mathrm{GL}(V_l)$$

acting on $\mathrm{Rep}_d(C_{L+1})$ via

$$(g \cdot \phi)_l := g_l \, \phi_l \, g_{l-1}^{-1}, \quad (l = 0, \dots, L), \text{ mod } L+1$$

Then, for all $g \in G_d$ and all $l = 0, \dots, L$,

$$d\mu_l(g \cdot \phi) = g_{l-1} \, d\mu_l(\phi) \, g_l^{-1}.$$

Proof. Expand $d\mu_l(g \cdot \phi)$ as a product of the maps $(g \cdot \phi)_i$. Each factor contributes a left multiplication by some g_{i-1} and a right multiplication by g_i^{-1}, and all intermediate terms cancel by associativity, leaving only g_{l-1} on the left and g_l^{-1} on the right. $\square$

Proposition 4.8 (Orbit decomposition of Γ_d)**.** Let

$$\Gamma_{d,L+1} := \Big\{\phi \in \mathrm{Rep}_d(C_{L+1}) \;\Big|\; d\mu_l(\phi) = 0 \text{ for all } l = 0, \dots, L\Big\}.$$

Then $\Gamma_{d,L+1}$ is G_d-invariant. In particular, $\Gamma_{d,L+1}$ decomposes as a disjoint union of G_d-orbits:

$$\Gamma_{d,L+1} = \bigsqcup_{[\phi] \in \Gamma_{d,L+1}/G_d} G_d \cdot \phi.$$

Proof. By Proposition 4.7, each map $d\mu_l$ is G_d-equivariant and hence the vanishing set $d\mu_l^{-1}(0)$ is G_d-stable. Intersecting over $l = 0, \dots, L$ shows that $\Gamma_{d,L+1}$ is G_d-stable. Any group action partitions a set into disjoint orbits, yielding the stated decomposition. $\square$

Informally, we have $\Gamma_{d,L+1} \cong \mathrm{Rep}_d(C_{L+1})/ < d\mu_l >$, we will that $\Gamma_{d,L+1}$ is then finite dimensional and we can classify indecomposable using indecomposable representations of Nakayama algebras.

4.2 Cyclic quiver as a bounded module

Definition 4.9 (Path algebra)**.** Let Q be a quiver. The *path algebra* kQ is the k-vector space with basis given by all directed paths in Q (including the trivial paths e_i of length 0 at each vertex i), with multiplication given by concatenation of composable paths and 0 otherwise, extended by bilinearity.

Definition 4.10 (Bound quiver algebra associated with Γ_d)**.** Let $Q = C_{L+1}$ and for each $l = 0, \dots, L$ let $p_l \in kQ$ be the path

$$p_l := a_{l-1} \cdots a_1\, a_0 \cdots a_{l+1}, \ \bmod L + 1$$

interpreting empty products as the corresponding trivial path. Let $I \subset kQ$ be the two-sided ideal generated by $\{p_0, \dots, p_L\}$. The quotient algebra

$$A := kQ/I$$

is called the *bound quiver algebra* associated with these relations.

Definition 4.11 (Representation of an algebra of dimension d)**.** Let A be a k-algebra and let $d = (d_0, \dots, d_L)$. A *representation of* A *of dimension* d *compatible with the vertices* is a left A-module structure on $V := \bigoplus_{i=0}^{L} k^{d_i}$ such that the idempotents $e_i \in kQ \twoheadrightarrow A$ act as the projections onto the i-th summand. We denote the corresponding representation space by $\mathrm{Rep}_d(A)$.

Proposition 4.12 (From quiver representations with relations to A-modules)**.** There is a well-defined map

$$F : \Gamma_{d,L+1} \to \mathrm{Rep}_d(A)$$

which sends $\phi \in \Gamma_{d,L+1}$ to the unique A-module structure on V for which each arrow a_i acts by the linear map ϕ_i.

Proof. Given $\phi \in \Gamma_{d,L+1}$, the tuple of linear maps $(\phi_i)_i$ defines a representation of the quiver $Q = C_{L+1}$, hence a left kQ-module structure on V by letting a path act by the corresponding composition of arrow maps. For each l, the defining condition $d\mu_l(\phi) = 0$ is exactly the statement that the element $p_l \in kQ$ acts as the zero map on V. Since I is the two-sided ideal generated by the p_l, every element of I also acts by zero. Therefore the kQ-action factors through the quotient $A = kQ/I$, producing a unique A-module structure on V with the required arrow actions. □

Proposition 4.13 (From A-modules to quiver representations with relations)**.** There is a well-defined map

$$G : \mathrm{Rep}_d(A) \to \Gamma_{d,L+1}$$

which sends an A-module structure on V to the induced quiver representation ϕ whose arrow maps are the actions of the classes of the arrows a_i.

Proof. Let V be a representation in $\mathrm{Rep}_d(A)$. Composing the action map $A \to \mathrm{End}_k(V)$ with the quotient map $kQ \twoheadrightarrow A$ gives a kQ-module structure on V. Restricting the action of each arrow basis element a_i to the appropriate vertex summand yields linear maps $\phi_i : V_{i-1} \to V_i$ and hence a point $\phi \in \mathrm{Rep}_d(C_{L+1})$. Because each p_l maps to zero in A, it acts as the zero endomorphism on V. Equivalently, the composed linear map $\mu_l(\phi)$ is zero for every l, so $\phi \in \Gamma_{d,L+1}$. □

Proposition 4.14 (Correspondence between $\Gamma_{d,L+1}$ and $\mathrm{Rep}_d(A)$)**.** The maps F and G are mutually inverse bijections. In particular,

$$\Gamma_{d,L+1} \cong \mathrm{Rep}_d(A), \qquad A = kQ/I.$$

Proof. Starting from $\phi \in \Gamma_{d,L+1}$, the A-module $F(\phi)$ is by construction generated by the same arrow actions as ϕ, so applying G recovers the same tuple of arrow maps, hence $G(F(\phi)) = \phi$. Conversely, starting from an A-module structure on V, the map G records the arrow actions and the map F reconstructs the unique A-module whose arrow actions agree with those recorded, hence $F(G(V)) = V$. □

Lemma 4.15 (Finite-dimensionality of $A = kQ/I$)**.** Let $Q = C_{L+1}$ be the oriented cycle on $L+1$ vertices, and let $I \subset kQ$ be the two-sided ideal generated by the paths $p_0, \ldots, p_L$, where each p_l is a path of length L (the unique directed path going around the cycle while omitting one arrow). Then every directed path in Q of length $\geq L+1$ lies in I. In particular, $A = kQ/I$ is finite-dimensional.

Proof. Let q be any directed path in Q of length $\geq L+1$. Write $q = b_m b_{m-1} \cdots b_1$ as a concatenation of arrows with $m \geq L+1$. Consider the contiguous subpath $q' := b_L \cdots b_1$ of length L. Since Q is an oriented $L+1$-cycle, there is exactly one directed path of length L from its source to its target, hence q' must coincide with one of the generators p_l. Therefore $q = q''\, p_l$ for some (possibly trivial) path q'', so $q \in I$ because I is a two-sided ideal.

Thus all paths of length $\geq L+1$ vanish in A, so A is spanned by the residue classes of paths of length $< L+1$, a finite set. Hence A is finite-dimensional. □

4.3 Bounded modules are Nakayama algebras

Definition 4.16 (A-module)**.** Let A be a (not necessarily commutative) k-algebra. A (left) A-*module* is a k-vector space M together with a k-bilinear map

$$A \times M \to M, \qquad (a, m) \mapsto a \cdot m,$$

such that for all $a, b \in A$ and $m \in M$ one has

$$(ab) \cdot m = a \cdot (b \cdot m), \qquad 1_A \cdot m = m.$$

A *homomorphism* of A-modules is a k-linear map commuting with the A-action.

Definition 4.17 (Indecomposable module)**.** An A-module M is *indecomposable* if $M \neq 0$ and there do not exist nonzero submodules $M_1, M_2 \subset M$ such that

$$M \cong M_1 \oplus M_2.$$

Equivalently, M is indecomposable if every idempotent endomorphism of M is either 0 or id_M.

Definition 4.18 (Composition series)**.** Let M be a finite-dimensional A-module. A *composition series* of M is a finite chain of submodules

$$0 = M_0 \subset M_1 \subset \cdots \subset M_\ell = M$$

such that each successive quotient M_i/M_{i-1} is *simple*. The integer ℓ is called the *length* of the series.

Definition 4.19 (Projective module)**.** An A-module P is *projective* if for every surjective A-module homomorphism $f : M \twoheadrightarrow N$ and every A-module homomorphism $g : P \to N$, there exists an A-module homomorphism $\tilde{g} : P \to M$ such that $f \circ \tilde{g} = g$. Equivalently, P is projective if the functor $\mathrm{Hom}_A(P, -)$ is exact.

Definition 4.20 (Indecomposable projective module)**.** A *projective indecomposable* A-module is an A-module that is both projective and indecomposable. For a finite-dimensional basic algebra A, every indecomposable projective module is of the form $P_i \cong Ae_i$ for a primitive idempotent $e_i \in A$, and every projective module decomposes uniquely as a direct sum of indecomposable projectives.

Definition 4.21 (Uniserial module)**.** An A-module M is *uniserial* if it admits a unique composition series. Equivalently, the set of submodules of M is totally ordered by inclusion.

Definition 4.22 (**Nakayama algebra**)**.** Let A be a finite-dimensional k-algebra. We say that A is a *Nakayama algebra* if every indecomposable left A-module is *uniserial*, i.e. it admits a unique composition series. Equivalently, A is Nakayama if every indecomposable projective left A-module is uniserial (and hence every indecomposable injective is uniserial).

Definition 4.23 (Admissible ideal)**.** Let Q be a finite quiver and let kQ be its path algebra. Write $(kQ)_{\geq m}$ for the k-span of all paths of length at least m. A two-sided ideal $I \subset kQ$ is called *admissible* if there exists an integer $m \geq 2$ such that

$$(kQ)_{\geq m} \subseteq I \subseteq (kQ)_{\geq 2}.$$

Definition 4.24 (Jacobson radical)**.** Let A be a (not necessarily commutative) ring. The *Jacobson radical* $J(A)$ is the intersection of all maximal left ideals of A. If A is a finite-dimensional k-algebra, then equivalently

$$J(A) = \bigcap_{S \text{ simple}} \ker\big(A \to \mathrm{End}_k(S)\big),$$

i.e. $J(A)$ is the largest two-sided ideal that acts by zero on every simple A-module.

Proposition 4.25 (**Bound quiver criterion for Nakayama algebras**)**.** Let k be a field, let $A \simeq kQ/I$ be a finite-dimensional bound quiver algebra, where Q is a finite quiver and $I \subset kQ$ is an admissible ideal. Assume that for every vertex $i \in Q_0$ there is at most one arrow leaving i and at most one arrow entering i. Then A is a Nakayama algebra.

Moreover, if Q is connected and has exactly one arrow leaving and exactly one arrow entering each vertex, then Q is an oriented cycle and A is a *cyclic Nakayama algebra*. If instead Q is connected and has a unique source and a unique sink, then Q is an oriented line and A is a *linear Nakayama algebra*.

Proof. Let $e_i \in kQ \twoheadrightarrow A$ be the primitive idempotent corresponding to a vertex $i \in Q_0$, and consider the indecomposable projective module $P_i := Ae_i$. As a k-vector space, P_i is spanned by residue classes of paths in Q starting at i (modulo the relations in I). Because there is at most one arrow leaving each vertex, for each $\ell \geq 0$ there is *at most one* directed path of length ℓ starting at i; hence P_i has a basis indexed by (the surviving classes of) these paths.

Let $J \subset A$ denote the Jacobson radical. Since $A \simeq kQ/I$ with I admissible, J is generated (as a two-sided ideal) by the classes of arrows, and one has

$$J^\ell P_i = J^\ell A e_i \;\cong\; (J^\ell)e_i,$$

which is spanned by classes of paths of length $\geq \ell$ starting at i. By the previous paragraph, for each ℓ the quotient $J^\ell P_i / J^{\ell+1} P_i$ is either 0 or a simple module (it is spanned by the class of the unique surviving path of length ℓ, if it exists). Therefore the radical filtration

$$P_i \supset JP_i \supset J^2 P_i \supset \cdots$$

is a chain whose successive quotients are simple (or zero), i.e. P_i is uniserial.

Thus every indecomposable projective P_i is uniserial, and hence A is a Nakayama algebra.

For the final statements: if Q is connected and every vertex has exactly one incoming and one outgoing arrow, then the underlying directed graph is a directed cycle. If Q is connected and has a unique source and sink with all other vertices having one incoming and one outgoing arrow, then the directed graph is a directed line. These correspond to cyclic and linear Nakayama algebras respectively. □

4.4 Cyclic monomial Nakayama algebra with uniform Kupisch series

Throughout this subsection, let $Q = C_{L+1}$ be the oriented $L+1$-cycle, let kQ be its path algebra, and let

$$A := kQ/I$$

be the bound quiver algebra from Definition (Bound quiver algebra associated with $\Gamma_{d,L+1}$).

Definition 4.26 (Monomial (path) ideal / monomial algebra)**.** An ideal $I \subset kQ$ is called *monomial* if it is generated (as a two-sided ideal) by paths in Q. A quotient kQ/I with I monomial is called a *monomial algebra*.

Definition 4.27 (Kupisch series)**.** Let A be a finite-dimensional basic algebra with primitive idempotents $e_0, \ldots, e_L$ and indecomposable projectives $P_i := Ae_i$. The *Kupisch series* of A is the $L+1$-tuple

$$(c_0, \ldots, c_L), \qquad c_i := \ell(P_i),$$

where $\ell(P_i)$ denotes the composition length of P_i.

Remark 4.28 (Loewy length)**.** For a finite-dimensional algebra A with Jacobson radical $J = J(A)$, the *Loewy length* is the smallest integer n such that $J^n = 0$.

Proposition 4.29 (Identification and Kupisch series)**.** With $Q = C_{L+1}$ and $I = (p_0, \dots, p_L)$ as above, the algebra $A = kQ/I$ is a *cyclic monomial Nakayama algebra.* Moreover its Kupisch series is uniform:

$$(c_0, \dots, c_L) = (L, \dots, L).$$

Proof. By construction, the ideal I is generated by the paths $p_0, \dots, p_L$, hence I is monomial and A is a monomial algebra. By Proposition 4.25, A is Nakayama; since $Q = C_{L+1}$ is an oriented cycle, A is in fact *cyclic* Nakayama.

Fix a vertex i and consider $P_i = Ae_i$. Modulo I, every path in Q of length $\geq L$ starting at i vanishes, while for each $t = 0, 1, \dots, L-1$ there is exactly one directed path of length t starting at i. Therefore P_i has a k-basis given by the residue classes of these L paths (length 0 through $L-1$).

Since $J = J(A)$ is generated by the classes of arrows (as recalled in the proof of Proposition 4.25), the radical filtration

$$P_i \supset JP_i \supset J^2 P_i \supset \cdots$$

has successive quotients $J^t P_i / J^{t+1} P_i$ one-dimensional (hence simple) for $t = 0, \dots, L-1$, and zero thereafter. Thus $\ell(P_i) = L$ for every i, which gives the Kupisch series $(L, \dots, L)$. ☐

Remark 4.30. Equivalently, A can be viewed as a *truncated cycle algebra*: it is the path algebra of an oriented cycle with all paths of length $\geq L$ killed.

4.5 Indecomposable representations of Nakayama algebras

Definition 4.31 (Interval modules)**.** Fix $Q = C_{L+1}$ with vertices $Q_0 = \{0, \dots, L\}$ and arrows $a_t : t-1 \to t$ for $t = 1, \dots, L$, together with $a_0 : L \to 0$. The *interval module* $M(i, \ell)$, starting at vertex i and of length ℓ, is the representation of Q defined by

$$M(i,\ell)_j := \begin{cases} k, & j \in \{i, i+1, \dots, i+\ell\} \pmod{L+1}, \\ 0, & \text{otherwise,} \end{cases}$$

and for each arrow $a_t : t-1 \to t$,

$$M(i,\ell)_{a_t} := \begin{cases} \mathrm{id}_k, & t \in \{i+1, \dots, i+\ell\} \pmod{L+1}, \\ 0, & \text{otherwise.} \end{cases}$$

Lemma 4.32 (The relations vanish on $M(i, \ell)$)**.** Let $Q = C_{L+1}$ and, for each $l \in \{0, \dots, L\}$, let

$$p_l := a_{l-1} \cdots a_1\, a_0 a_L \cdots a_{l+1}, \quad \mod L+1$$

be the oriented path of length L going once around the cycle and omitting a_l. Set $A := kQ/\langle p_0, \dots, p_L \rangle$. Then, for every $i \in Q_0$ and $0 \leq \ell \leq L$, the representation $M(i, \ell)$ factors through A, hence defines a finite-dimensional left A-module.

Proof. Fix i and ℓ. Each path p_l has length L. Since the support of $M(i, \ell)$ contains only $\ell \leq L$ consecutive vertices, the corresponding composition along p_l necessarily passes through a vertex where $M(i, \ell)$ is zero. Therefore p_l acts by the zero map on $M(i, \ell)$ for every l, so all generators of the ideal $\langle p_0, \dots, p_L \rangle$ act trivially. Hence the kQ-action factors through A. ☐

Lemma 4.33 (Indecomposability of $M(i,\ell)$)**.** For every interval module $M(i,\ell)$, the A-module $M(i,\ell)$ is uniserial, and in particular indecomposable.

Proof. Let $N \subseteq M(i,\ell)$ be a nonzero submodule. Since $N \neq 0$, there exists a vertex t such that $N_t \neq 0$. As $N_t \subseteq M(i,\ell)_t$ and each nonzero vertex space of $M(i,\ell)$ is one-dimensional, we must have

$$N_t = M(i,\ell)_t \cong k.$$

Write the support of $M(i,\ell)$ as the ordered interval

$$i,\ i+1,\ \dots,\ i+\ell$$

(along the chosen orientation, modulo $L+1$), and choose t minimal such that $N_t \neq 0$. Then $N_s = 0$ for all vertices s preceding t in this interval. Now for each vertex r with

$$t \leq r \leq i+\ell-1,$$

the arrow

$$a_{r+1} \colon r \to r+1$$

acts as the identity on $M(i,\ell)$. Since N is a subrepresentation and $N_r = M(i,\ell)_r \cong k$, its image under a_{r+1} is nonzero, hence

$$N_{r+1} = M(i,\ell)_{r+1}.$$

By induction we obtain

$$N_r = M(i,\ell)_r \qquad \text{for all } r = t, t+1, \dots, i+\ell,$$

and $N_r = 0$ at all earlier vertices in the support and outside the support. Therefore every nonzero submodule N is exactly a terminal interval submodule, namely

$$N \cong M\big(t,\ i+\ell-t\big)$$

(with indices read modulo $L+1$). Hence the nonzero submodules of $M(i,\ell)$ are precisely the terminal subintervals of the support, and these are totally ordered by inclusion. Explicitly, they form the chain

$$0 \subset M(i+\ell,0) \subset M(i+\ell-1,1) \subset M(i+\ell-2,2) \subset \cdots \subset M(i,\ell).$$

Here $M(i+\ell-r,r)$ is the submodule supported on the vertices

$$i+\ell-r,\ i+\ell-r+1,\ \dots,\ i+\ell,$$

that is, the terminal interval of length r inside the support

$$i,\ i+1,\ \dots,\ i+\ell.$$

Thus $M(i,\ell)$ is uniserial. Finally, every uniserial module is indecomposable: if

$$M(i,\ell) = U \oplus V$$

with $U, V \neq 0$, then U and V are two nonzero submodules neither of which contains the other, contradicting the fact that the submodules are totally ordered. Hence $M(i,\ell)$ is indecomposable. $\square$

Lemma 4.34 (Endomorphisms of $M(i,\ell)$)**.** For every interval module $M(i,\ell)$ one has

$$\mathrm{End}_A(M(i,\ell)) \cong k.$$

Proof. Let $f \in \mathrm{End}_A(M(i,\ell))$. At each vertex t in the support $[i \rightsquigarrow i+\ell]$, the map f_t is multiplication by some scalar $\lambda_t \in k$. For each arrow inside the support, the arrow map is id_k, so commutativity of

$$f_{t+1} \circ M(i,\ell)_{a_{t+1}} = M(i,\ell)_{a_{t+1}} \circ f_t$$

implies $\lambda_{t+1} = \lambda_t$. Hence all λ_t coincide, and therefore $f = \lambda\,\mathrm{id}$ for some $\lambda \in k$. □

§

Lemma 4.35 (Projective truncations realize interval modules)**.** Let $P(i) := Ae_i$ be the indecomposable projective at vertex i and let $J := J(A)$. Then, for every $i \in Q_0$ and every $0 \le \ell \le L-1$, there is an isomorphism of A-modules

$$P(i)/J^{\ell+1}P(i) \cong M(i,\ell),$$

where indices are taken modulo $L+1$.

Proof. By Proposition 4.25 and the previous subsection, A is a cyclic Nakayama algebra with uniform Kupisch series $(L,\dots,L)$. In particular, $P(i)$ is uniserial of length L, and $J^tP(i)$ is spanned by residue classes of paths of length $\ge t$ starting at i. Thus $P(i)/J^{\ell+1}P(i)$ has a basis given by the residue classes of the unique paths of lengths

$$0,1,\dots,\ell$$

starting at i. Under the induced quiver description, this quotient has one-dimensional spaces on the vertices

$$i, i+1, \dots, i+\ell \pmod{L+1},$$

and identity maps along arrows inside the support. This is precisely the interval module $M(i,\ell)$. □

Proposition 4.36 (Indecomposables of the cyclic truncated Nakayama algebra)**.** Let $A = kQ/I$ be the cyclic truncated Nakayama algebra associated with $Q = C_{L+1}$, where I is generated by the paths $p_0,\dots,p_L$ of length L. Then every finite-dimensional indecomposable A-module is isomorphic to an interval module

$$M(i,\ell)$$

for a unique pair (i,ℓ) such that

$$i \in Q_0, \qquad 0 \le \ell \le L-1.$$

Equivalently, every indecomposable A-module is of the form

$$P(i)/J^{\ell+1}P(i)$$

for a unique pair (i,ℓ) with

$$0 \le \ell \le L-1.$$

In particular, A has exactly

$$L(L+1)$$

indecomposable modules up to isomorphism.

Proof. By Lemma 4.32, each $M(i, \ell)$ is a well-defined finite-dimensional A-module, and by Lemma 4.33 it is indecomposable. Conversely, since A is Nakayama (Proposition 4.25), every indecomposable A-module is uniserial and hence a quotient of a unique indecomposable projective $P(i) = Ae_i$. Because all paths of length L vanish in A, the indecomposable projective $P(i)$ has composition length L. Hence every indecomposable quotient of $P(i)$ is of the form

$$P(i)/J^{\ell+1}P(i)$$

for some $0 \leq \ell \leq L-1$. By Lemma 4.33, this module is isomorphic to

$$M(i, \ell).$$

Uniqueness follows because the top simple module determines i and the length determines ℓ. For each $i \in Q_0$ there are L possible values of ℓ, namely $0, \dots, L-1$. Since $|Q_0| = L+1$, this yields a total of

$$L(L+1)$$

indecomposable modules up to isomorphism. □

More details about Nakayama algebras and their representations can be found in the following lecture notes: Noncommutative Algebra 2: Representations of finite-dimensional algebras in particular theorem and corollary section 1.6. We have a Krull–Schmidt theorem for these Nakayama algebras.

Proposition 4.37 (Krull–Schmidt for the cyclic truncated Nakayama algebra)**.** Let $A = kQ/I$ be the cyclic truncated Nakayama algebra, where $Q = C_{L+1}$ is the oriented $(L+1)$-cycle and I is the ideal generated by the paths $p_0, \dots, p_L$ of length L (equivalently, all paths of length $\geq L$ vanish in A). Let mod-A denote the category of finite-dimensional left A-modules. Then every $M \in$ mod-A admits a decomposition as a finite direct sum of indecomposable A-modules. Moreover, this decomposition is unique up to reordering and isomorphism of the indecomposable summands. Equivalently, using the explicit indecomposables $M(i, \ell)$ from proposition 4.36, there exist uniquely determined multiplicities $m_{i,\ell} \in \mathbb{Z}_{\geq 0}$ such that

$$M \;\cong\; \bigoplus_{i=0}^{L} \bigoplus_{\ell=0}^{L-1} M(i, \ell)^{\oplus m_{i,\ell}}.$$

Proof sketch. Since A is finite-dimensional over k, it is left artinian (and left noetherian), hence every object of mod-A has finite length. In particular, mod-A is a Krull–Schmidt category: every object decomposes as a finite direct sum of indecomposables, and such a decomposition is unique up to permutation and isomorphism. Concretely, existence follows by choosing a nonzero module M and repeatedly splitting off a nontrivial direct summand until all summands are indecomposable; the process terminates because M has finite length. Uniqueness follows because each indecomposable has a local endomorphism ring (for example, by Lemma 4.34 in the explicit classification), and the classical Krull–Schmidt–Azumaya argument applies. Finally, Proposition 4.36 classifies the indecomposables as the modules $M(i, \ell)$ with $0 \leq \ell \leq L-1$, so any Krull–Schmidt decomposition can be rewritten uniquely in the stated multiplicity form. □

Note: This latter proposition enables to parameterize the orbits of Γ with the indecomposables representations.

Definition 4.38. Consider a A module M and its Krull-Schmidt decomposition into $M(i,\ell)$ for $i \in \{0,\dots,L\}$ and $\ell \in \{0,\dots,L-1\}$. For each vertex k the dimension d_k of the representation M at vertex k satisfies the dimension constraint:

$$d_k \;=\; \sum_{i=0}^{L}\sum_{\ell=0}^{L-1} m_{i,\ell}\,\mathbf{1}\{k \in \{i, i+1, \dots, i+\ell\} \pmod{L+1}\} \qquad k = 0,\dots,L$$

Define the set of *multiplicity patterns* as

$$\mathcal{M}_d^+ := \left\{ (m_{i,\ell}) \in \mathbb{Z}_{\geq 0}^{L(L+1)} \,\middle|\, \text{the above dimension constraints hold for all } k \right\}.$$

Definition 4.39. For each $m \in \mathcal{M}_d^+(A)$, define the stratum

$$\mathcal{O}_m := \left\{ \phi \in \Gamma_{d,L+1} \,\middle|\, M(\phi) \cong \bigoplus_{i,\ell} M(i,\ell)^{\oplus m_{i,\ell}} \right\},$$

$\mathcal{O}_m$ is a single G_d-orbit and ϕ_m is a particular representative in this orbit.

Proposition 4.40 (Orbit stratification of $\Gamma_{d,L+1}$ by indecomposable Nakayama modules)**.** Fix $L \geq 1$ and let $Q = C_{L+1}$ be the oriented cyclic quiver on vertices $\{0,\dots,L\}$. Let $A := kQ/I$ be the bound quiver algebra, where I is the ideal generated by the length-L paths $p_0,\dots,p_L$ obtained by going once around the cycle and omitting one arrow. Let $d = (d_0,\dots,d_L)$ be a dimension vector. The variety $\Gamma_{d,L+1}$ decomposes as a disjoint union of these orbits:

$$\Gamma_{d,L+1} \;=\; \bigsqcup_{m \in \mathcal{M}_d^+(A)} \mathcal{O}_m \;=\; \bigsqcup_{m \in \mathcal{M}_d^+(A)} G_d \cdot \phi_m,$$

where ϕ_m is any representative whose associated A-module has multiplicity pattern m. In particular, the G_d-orbits in $\Gamma_{d,L+1}$ are parameterized by the multiplicity data of indecomposable A-modules, i.e. by the set $\mathcal{M}_d^+(A)$.

5 Stratification of vanishing cyclic conditions with fixed target map

While the indecomposable representation of $\Gamma_{d,\sigma}$ are parameterized by the $M(i,\ell)$ we are actually interested in stratifying:

$$\Gamma_{d,\sigma} := \{(\phi_0,\dots,\phi_L) \in \mathrm{Rep}_d(C_{L+1}) \mid \phi_{l-1}\cdots\phi_1\,\phi_0\,\phi_L\cdots\phi_{l+1} = 0,\ \phi_0 = \sigma\}\,.$$

where the map $\sigma \in \mathrm{Hom}(V_L, V_0)$ is fixed (for a DLN it is the map $\Sigma_{XY}U_Q$). The map

$$d\mu_{l,\sigma} := \phi_{l-1}\cdots\phi_1\,\sigma\,\phi_L\cdots\phi_{l+1}$$

is no longer G_d-equivariant.

Definition 5.1 (Stabilizer group of the fixed arrow)**.** Let $\sigma \in \mathrm{Hom}(V_L, V_0)$ be fixed. Define

$$\mathrm{Stab}(\sigma) := \{(g_0, g_L) \in \mathrm{GL}(V_0) \times \mathrm{GL}(V_L) \mid g_0\, \sigma\, g_L^{-1} = \sigma\},$$

and the corresponding subgroup of the base–change group

$$G_\sigma := \{g = (g_0, \dots, g_L) \in \prod_{l=0}^{L} \mathrm{GL}(V_l) \mid (g_0, g_L) \in \mathrm{Stab}(\sigma)\}.$$

Proposition 5.2 (G_σ-equivariance of the cyclic composition maps)**.** Fix $\sigma \in \mathrm{Hom}(V_L, V_0)$ and consider the representation space with $\phi_0 = \sigma$. For $\phi = (\sigma, \phi_1, \dots, \phi_L)$ and each $l = 0, \dots, L$, define

$$d\mu_{l,\sigma}(\phi) := \phi_{l-1} \cdots \phi_1\, \sigma\, \phi_L \cdots \phi_{l+1} \in \mathrm{Hom}(V_l, V_{l-1}),$$

with the convention that an empty product is the identity. Then, for all $g \in G_\sigma$ and all l,

$$d\mu_{l,\sigma}(g \cdot \phi) = g_l\, d\mu_{l,\sigma}(\phi)\, g_{l+1}^{-1}.$$

In particular, the subset $\Gamma_{d,\sigma} = \bigcap_{l=0}^{L} d\mu_{l,\sigma}^{-1}(0)$ is G_σ-invariant.

Proof. Expand $d\mu_{l,\sigma}(g \cdot \phi)$ as a product of the maps $(g \cdot \phi)_i$. All intermediate base-change factors cancel. The only remaining factors are g_l on the left and g_{l+1}^{-1} on the right. The middle arrow is $(g \cdot \phi)_0 = g_0 \sigma g_L^{-1}$, which equals σ because $g \in G_\sigma$. $\square$

For all modules in $\Gamma_{d,\sigma}$ Krull-Schmidt applies (as a category of A-modules where A is finite dimensional). We want to parameterize the orbits of $\Gamma_{d,\sigma}$ under the action of G_σ to obtain a combinatorial description of the saddle points.

Lemma 5.3 (Intersection of a G_d–orbit with a fixed–arrow fiber)**.** Let $L \geq 2$ and let $V_0, \dots, V_L$ be finite–dimensional vector spaces over a field $\Bbbk$. Let $\mathcal{O} \subset \Gamma_{d,L+1}$ be a single G_d–orbit. Then either $\mathcal{O} \cap \Gamma_{d,\sigma} = \varnothing$, or else for every $x \in \mathcal{O} \cap \Gamma_{d,\sigma}$ one has

$$\mathcal{O}_\sigma := \mathcal{O} \cap \Gamma_{d,\sigma} \;=\; G_\sigma \cdot x.$$

In particular, whenever $\mathcal{O} \cap \Gamma_{d,\sigma} \neq \varnothing$, this intersection consists of a single G_σ–orbit.

We prove this lemma in the appendix 5.3.

Proposition 5.4 (Rank label for Nakayama orbits and the fiber non-emptiness criterion over $\mathbb{R}$)**.** Consider the morphism:

$$F : \Gamma_d \longrightarrow \mathrm{Hom}_{\mathbb{R}}(V_L, V_0)$$

satisfying the equivariance identity

$$F\big((g_0, \dots, g_L) \cdot x\big) = g_0\, F(x)\, g_L^{-1} \qquad \text{for all } (g_0, \dots, g_L) \in G_d,\ x \in \Gamma_d. \tag{7}$$

Consider a stratification of $\Gamma_{d,\sigma}$ into G_d–orbits indexed by a parameter set Λ,

$$\Gamma_{d,\sigma} \;=\; \bigsqcup_{\lambda \in \Lambda} \mathcal{O}_\lambda,$$

where $\mathcal{O}_\lambda$ denotes the G_d–orbit corresponding to λ (in the context of this paper, λ corresponds to the Krull–Schmidt multiplicities of the Nakayama indecomposables). For each $\lambda \in \Lambda$, define the integer

$$r(\lambda) \;:=\; \operatorname{rank}\bigl(F(x)\bigr) \quad \text{for any } x \in \mathcal{O}_\lambda. \tag{8}$$

Then the following statements hold.

(i) The integer $r(\lambda)$ in (8) is well-defined, i.e. it does not depend on the choice of $x \in \mathcal{O}_\lambda$.

(ii) Fix $\sigma \in W$ and define the fiber $\Gamma_\sigma := F^{-1}(\sigma)$. Then for every $\lambda \in \Lambda$,

$$\mathcal{O}_\lambda \cap \Gamma_\sigma \neq \varnothing \qquad \Longleftrightarrow \qquad r(\lambda) = \operatorname{rank}(\sigma). \tag{9}$$

The proof of this proposition is given in the appendix 5.3.

Definition 5.5. Fix $L \geq 2$ and a dimension vector $d = (d_0, \ldots, d_L)$. Recall that $\Gamma_{d,L+1}$ decomposes into G_d–orbits indexed by $\mathcal{M}_d^+$, i.e. for each

$$\mathbf{m} = (m_{i,\ell}) \in \mathcal{M}_d^+$$

there is a corresponding G_d–orbit $\mathcal{O}_{\mathbf{m}} \subset \Gamma_{d,L+1}$ such that every $\phi \in \mathcal{O}_{\mathbf{m}}$ has associated Nakayama module

$$M(\phi) \;\cong\; \bigoplus_{i,\ell} M(i,\ell)^{\oplus m_{i,\ell}}$$

(up to isomorphism), where $M(i,\ell)$ denote the indecomposable modules of the cyclic Nakayama algebra. For $\mathbf{m} \in \mathcal{M}_d^+$ define

$$r(\mathbf{m}) \;:=\; \operatorname{rank}(F(\phi)) \qquad \text{for any } \phi \in \mathcal{O}_{\mathbf{m}},$$

which is well-defined by Proposition 5.4.

Definition 5.6. For an indecomposable $M(i,\ell)$, define its *support interval* to be the cyclic set of vertices

$$\operatorname{Supp}(i,\ell) \;:=\; \{\, i,\; i+1,\; \ldots,\; i+\ell \,\} \subset \mathbb{Z}/(L+1)\mathbb{Z},$$

where addition is taken modulo $L+1$ and we identify vertices with their residue classes. Say that $M(i,\ell)$ is $(L \to 0)$*–active* if the arrow $L \to 0$ acts nontrivially on $M(i,\ell)$, i.e. if the closing map

$$M(i,\ell)(L) \longrightarrow M(i,\ell)(0)$$

induced by the quiver arrow $L \to 0$ is nonzero. Define the indicator

$$\epsilon(i,\ell) \;:=\; \begin{cases} 1, & \text{if } M(i,\ell) \text{ is } (L \to 0)\text{–active}, \\ 0, & \text{otherwise.} \end{cases}$$

Proposition 5.7 (Rank formula for the closing arrow in terms of Nakayama multiplicities)**.** For every $\mathbf{m} = (m_{i,\ell}) \in \mathcal{M}_d^+$ one has the formula

$$r(\mathbf{m}) \;=\; \sum_{i=0}^{L} \sum_{\ell=1}^{L} \epsilon(i,\ell)\, m_{i,\ell}. \tag{10}$$

The proof of that proposition is given in the appendix 5.3.

Corollary 5.8 (Orbit decomposition of Γ_{XY} indexed by $\mathcal{M}_d^+$)**.** Recall $\Gamma_{d,XY} = \Gamma_{d,\sigma} := F^{-1}(\sigma)$, the G_d–orbits in Γ_d are denoted $\mathcal{O}_{\mathbf{m}}$ for $\mathbf{m} \in \mathcal{M}_d^+$, and $G_\sigma := \{g \in G_d : \ g_0 \sigma g_L^{-1} = \sigma\}$. Consider the orbit:

$$\mathcal{O}_{\mathbf{m},\sigma} := \mathcal{O}_{\mathbf{m}} \cap \Gamma_{d,XY}.$$

Then $\Gamma_{d,XY}$ decomposes as a disjoint union of G_σ–orbits indexed by those $\mathbf{m} \in \mathcal{M}_d^+$ satisfying the rank constraint:

$$\Gamma_{d,XY} \; = \bigsqcup_{\mathbf{m}\in\mathcal{M}_d^+ \,:\, r(\mathbf{m})=\mathrm{rank}(\sigma)} \mathcal{O}_{\mathbf{m},\sigma},$$

and for each such $\mathbf{m}$ the set $\mathcal{O}_{\mathbf{m},\sigma}$ is a single G_σ–orbit. Moreover, for $\mathbf{m} = (m_{i,\ell}) \in \mathcal{M}_d^+$ the integer $r(\mathbf{m})$ is given by the explicit formula

$$r(\mathbf{m}) \; = \; \sum_{i,\ell} \epsilon(i,\ell)\, m_{i,\ell},$$

where $\epsilon(i,\ell) \in \{0,1\}$ is the $(L \to 0)$–activity indicator of the indecomposable $M(i,\ell)$ as in Proposition 5.7.

Proof. Fix $\sigma \in \mathrm{Hom}(V_L, V_0)$ and recall $\Gamma_{d,XY} = \Gamma_{d,\sigma}$. First, since $\Gamma_{d,L+1} = \bigsqcup_{\mathbf{m}\in\mathcal{M}_d^+} \mathcal{O}_{\mathbf{m}}$ is a disjoint union, intersecting with Γ_σ gives the disjoint union

$$\Gamma_{d,\sigma} \; = \; \Gamma_{d,\sigma} \cap \Gamma_{d,L+1} \; = \; \Gamma_{d,\sigma} \cap \Gamma_{d,L+1} \; = \; \Gamma_{d,\sigma} \cap \left(\bigsqcup_{\mathbf{m}\in\mathcal{M}_d^+} \mathcal{O}_{\mathbf{m}} \right) \; = \; \bigsqcup_{\mathbf{m}\in\mathcal{M}_d^+} \mathcal{O}_{\mathbf{m},\sigma}.$$

By Proposition 5.4, for a given $\mathbf{m}$ the intersection $\mathcal{O}_{\mathbf{m}} \cap \Gamma_{d,\sigma}$ is nonempty if and only if $r(\mathbf{m}) = \mathrm{rank}(\sigma)$. Therefore all terms with $r(\mathbf{m}) \neq \mathrm{rank}(\sigma)$ vanish, and we obtain

$$\Gamma_{d,\sigma} \; = \bigsqcup_{\mathbf{m}\in\mathcal{M}_d^+ \,:\, r(\mathbf{m})=\mathrm{rank}(\sigma)} \mathcal{O}_{\mathbf{m},\sigma}.$$

Again by Proposition 5.4, whenever the intersection is nonempty, $\mathcal{O}_{\mathbf{m}} \cap \Gamma_{d,\sigma}$ is a single G_σ–orbit. This proves the stated orbit decomposition. $\square$

Remark 5.9 (Parametrization of G_σ–orbits in Γ_{XY})**.** The previous corollary can be equivalently phrased as follows. The G_σ–orbits in $\Gamma_{d,XY}$ are in bijection with the subset

$$\left\{ \mathbf{m} \in \mathcal{M}_d^+ \;\middle|\; r(\mathbf{m}) = \mathrm{rank}(\sigma) \right\},$$

via the correspondence

$$\mathbf{m} \; \longmapsto \; \mathcal{O}_{\mathbf{m}} \cap \Gamma_{d,\sigma}.$$

In particular, $\Gamma_{d,XY}$ decomposes as a disjoint union of G_σ–orbits indexed by Nakayama indecomposable multiplicity data satisfying the rank constraint $r(\mathbf{m}) = \mathrm{rank}(\sigma)$.

5.1 Cyclic rank patterns and multiplicities

Convention on interval modules. From this subsection onward we index indecomposable interval modules by their *endpoints*. Thus $M(k,\ell)$ denotes the indecomposable representation supported on the cyclic interval

$$[k \rightsquigarrow \ell],$$

with length $\delta(k,\ell) \leq L$. In particular the representation has one–dimensional vector spaces along the vertices $k, k+1, \ldots, \ell$ and identity maps along the arrows in this segment, and zero elsewhere. This convention differs from the length indexing used in previous sections and will be used from now on.

Cyclic intervals. For $i, j \in \{0, \ldots, L\}$ we denote by

$$[i \rightsquigarrow j]$$

the oriented segment obtained by following the arrows forward from i to j. Its length (number of arrows) is

$$\delta(i,j) \in \{0, \ldots, L\},$$

characterized by $j \equiv i + \delta(i,j) \pmod{L+1}$. Thus

$$[i \rightsquigarrow j] = \big(i \xrightarrow{\phi_{i+1}} i+1 \xrightarrow{\phi_{i+2}} \cdots \xrightarrow{\phi_j} j\big).$$

Cyclic rank patterns. Given $\phi = (\phi_0, \ldots, \phi_L) \in \Gamma_d$, for any pair (i,j) with $\delta(i,j) \leq L$ we define the composed path map

$$\Phi_{i\to j}(\phi) := \phi_j \phi_{j-1} \cdots \phi_{i+1} : V_i \longrightarrow V_j,$$

where indices are taken modulo $L+1$ and the product is taken along the segment $[i \rightsquigarrow j]$.

Definition 5.10. The *cyclic rank pattern* of ϕ is the collection of integers

$$r_{ij}(\phi) := \operatorname{rank}\big(\Phi_{i\to j}(\phi)\big), \qquad (i,j) \in \{0, \ldots, L\}^2 \text{ with } \delta(i,j) \leq L.$$

These ranks are invariant under the base–change action of $G_d = \prod_{k=0}^{L} GL(V_k)$ and therefore are constant on G_d–orbits.

Indecomposables and multiplicities. Let $m_{k\ell}$ denote the multiplicity of the indecomposable interval module $M(k,\ell)$ in ϕ. On $M(k,\ell)$ the map $\Phi_{i\to j}$ has rank 1 precisely when

$$[i \rightsquigarrow j] \subseteq [k \rightsquigarrow \ell],$$

and has rank 0 otherwise. Since ϕ decomposes as a direct sum of such interval modules, ranks add over summands and we obtain

$$r_{ij} = \sum_{\substack{k,\ell \\ [i\rightsquigarrow j]\subseteq[k\rightsquigarrow\ell]}} m_{k\ell}. \tag{11}$$

Explicit form. Fix i and write $j = i + t$ with $t = \delta(i, j) \in \{0, \dots, L\}$. An interval $[k \rightsquigarrow \ell]$ contains $[i \rightsquigarrow i + t]$ if and only if the vertices occur in the cyclic order

$$k \rightsquigarrow i \rightsquigarrow i + t \rightsquigarrow \ell.$$

Equivalently, the cyclic distances satisfy

$$\delta(k, i) + t + \delta(i + t, \ell) = \delta(k, \ell).$$

Hence

$$r_{i,\, i+t} = \sum_{\substack{k,\ell \\ k \rightsquigarrow i \rightsquigarrow i+t \rightsquigarrow \ell}} m_{k\ell}, \qquad t = 0, \dots, L. \tag{12}$$

Inversion formula. We now prove that the multiplicities m_{ij} can be recovered from the rank data r_{ij} by inclusion–exclusion.

Proposition 5.11. For every pair (i, j) with $\delta(i, j) \leq L - 1$, one has

$$m_{ij} = r_{ij} - r_{i-1,\, j} - r_{i,\, j+1} + r_{i-1,\, j+1}, \tag{13}$$

where indices are taken modulo $L + 1$, and by convention $r_{ab} = 0$ whenever $\delta(a, b) \geq L$.

The proof of this proposition is given in the appendix 5.3.

5.2 Cyclic rank patterns and the bijection $M_d^+ \simeq R_d^{\mathrm{orb}}$

Definition 5.12. For $\phi \in \Gamma_{d,L+1}$ we define its *cyclic rank pattern* to be the array

$$r(\phi) = (r_{ij}(\phi))_{0 \leq i,j \leq L} \qquad \text{where} \qquad r_{ij}(\phi) := \operatorname{rank}\bigl(\Phi_{ij}(\phi)\bigr).$$

Note that $r_{ii}(\phi) = \operatorname{rank}(\mathrm{id}_{V_i}) = d_i$. We now define a set of rank patterns and an "orbit-realizable" subset of that set.

Definition 5.13. Let R_d be the set of all integer arrays $r = (r_{ij})_{0 \leq i,j \leq L}$ such that

$$r_{ii} = d_i \qquad \text{for all } i \in \{0, \dots, L\}.$$

Define

$$R_d^{\mathrm{orb}} := \Bigl\{ r \in R_d \;\Bigm|\; r_{ij} - r_{i-1,\, j} - r_{i,\, j+1} + r_{i-1,\, j+1} \geq 0 \text{ for all } i \text{ and } \ell = 0, \dots, L \Bigr\},$$

where indices are taken modulo $L + 1$.

Lemma 5.14 (Forward formula)**.** Fix $m = (m_{ab}) \in M_d^+$, where m_{ab} is the multiplicity of the indecomposable interval module $M(a, b)$ supported on the oriented segment $[a \rightsquigarrow b]$, with $\delta(a, b) \leq L - 1$. For each pair i, j with $\delta(i, j) \leq L - 1$, define

$$r_{ij}(m) := \sum_{\substack{a,b \\ [i \rightsquigarrow j] \subseteq [a \rightsquigarrow b]}} m_{ab}.$$

Equivalently,

$$r_{ij}(m) = \sum_{u=0}^{\delta(i,j)} \sum_{v=0}^{L-\delta(i,j)-u-1} m_{\,i-u,\,j+v},$$

where all indices are taken modulo $L+1$. Equivalently again, $r_{ij}(m)$ is the sum of the multiplicities of all indecomposables whose support contains the segment $[i \rightsquigarrow j]$. If ϕ is any representation in the orbit corresponding to m, then

$$r_{ij}(m) = \operatorname{rank}\bigl(\Phi_{i\to j}(\phi)\bigr).$$

Hence $r(m) := (r_{ij}(m))$ depends only on the multiplicity pattern m, and not on the choice of ϕ in the corresponding orbit.

We prove lemma 5.14 in the appendix 5.3.

Lemma 5.15. For any $m \in M_d^+$, the rank pattern $r(m)$ lies in R_d^{orb}. Equivalently, for all pairs (i,j) with $\delta(i,j) \le L-1$,

$$r_{ij}(m) - r_{i-1,j}(m) - r_{i,j+1}(m) + r_{i-1,j+1}(m) \ge 0,$$

where, by convention, $r_{ab}(m) = 0$ whenever $\delta(a,b) \ge L$.

Proof. First, $r_{ii}(m)$ is the sum of the multiplicities of all indecomposables $M(a,b)$ whose support contains the vertex i. Hence $r_{ii}(m)$ is exactly the dimension at vertex i, so

$$r_{ii}(m) = d_i \qquad \text{for all } i.$$

Therefore $r(m) \in R_d$. Next, fix any pair (i,j) with $\delta(i,j) \le L-1$. By Proposition 5.11, applied to the rank pattern $r(m)$, we have

$$m_{ij} = r_{ij}(m) - r_{i-1,j}(m) - r_{i,j+1}(m) + r_{i-1,j+1}(m),$$

where, as usual, indices are taken modulo $L+1$, and the convention is that $r_{ab}(m) = 0$ whenever $\delta(a,b) \ge L$. Since $m \in M_d^+$, all multiplicities m_{ij} are nonnegative. Therefore

$$r_{ij}(m) - r_{i-1,j}(m) - r_{i,j+1}(m) + r_{i-1,j+1}(m) \ge 0$$

for every (i,j) with $\delta(i,j) \le L-1$. Thus $r(m)$ satisfies the defining inequalities of R_d^{orb}, and therefore

$$r(m) \in R_d^{\mathrm{orb}}.$$

□

Definition 5.16. Define

$$F : M_d^+ \longrightarrow R_d^{\mathrm{orb}}, \qquad F(m) := r(m).$$

Let $r \in R_d^{\mathrm{orb}}$. Define integers

$$m_{ij}(r) := r_{ij} - r_{i-1,\,j} - r_{i,\,j+1} + r_{i-1,\,j+1} \tag{14}$$

for all i,j, with indices modulo $L+1$ and the convention that $r_{i,\,j+1} = 0$ whenever the segment $[i \rightsquigarrow j+1]$ has length $\ge L$ (i.e. whenever $\delta(i,j+1) = L+1$ in the untruncated cycle; in practice, this means we set $r_{i,\,i+L+1} := 0$ to terminate the difference). Define:

$$G : R_d^{\mathrm{orb}} \longrightarrow M_d^+, \qquad G(r) := (m_{ij}(r))$$

Lemma 5.17. If $r \in R_d^{\mathrm{orb}}$, then $m_{ij}(r) \geq 0$ for all i, j, and $G(r) \in M_d^+$.

We proove this lemma in the appendix 5.3.

Proposition 5.18. The maps

$$F : M_d^+ \longrightarrow R_d^{\mathrm{orb}} \qquad \text{and} \qquad G : R_d^{\mathrm{orb}} \longrightarrow M_d^+$$

defined above are mutually inverse. Hence

$$M_d^+ \xrightarrow{\sim} R_d^{\mathrm{orb}}.$$

We prove this proposition in the appendix 5.3.

5.3 Orbit decomposition of $\Gamma_{d,XY}$ via cyclic rank patterns and the constraint $r(m) = \sigma$

Definition 5.19. For each $r \in R_d^{\mathrm{orb}}$ define the *σ-restricted orbit*

$$\mathcal{O}_{r,\sigma} := \mathcal{O}_r \cap \Gamma_{d,\sigma}.$$

Proposition 5.20. For every $r \in R_d^{\mathrm{orb}}$, the subset $\mathcal{O}_{r,\sigma}$ is either empty or a single G_σ-orbit. Furthermore, $\mathcal{O}_{r,\sigma} \neq \varnothing$ if and only if the scalar constraint

$$\mathrm{rank}(r) = \mathrm{rank}(\sigma)$$

holds, where

$$\mathrm{rank}(r) := \sum_{\substack{i,j \\ \delta(i,j) \geq 1}} \epsilon(i,j)\, m_{ij}(r) = \sum_{\substack{i,j \\ \delta(i,j) \geq 1}} \epsilon(i,j) \left(r_{ij} - r_{i-1,j} - r_{i,j+1} + r_{i-1,j+1} \right), \tag{15}$$

with

$$m_{ij}(r) := r_{ij} - r_{i-1,j} - r_{i,j+1} + r_{i-1,j+1},$$

indices taken modulo $L+1$, and the convention $r_{ab} = 0$ whenever $\delta(a,b) \geq L$. Here $\epsilon(i,j) \in \{0,1\}$ denotes the $(L \to 0)$-activity indicator of the indecomposable interval module $M(i,j)$.

We prove this proposition in the appendix 5.3.

Theorem 5.21 (Orbit stratification of $\Gamma_{d,\sigma}$ by cyclic rank patterns)**.** The algebraic variety $\Gamma_{d,\sigma}$ admits the disjoint union decomposition

$$\Gamma_{d,\sigma} = \bigsqcup_{\substack{r \in R_d^{\mathrm{orb}} \\ \mathrm{rank}(r) = \mathrm{rank}(\sigma)}} \mathcal{O}_{r,\sigma}. \tag{16}$$

In particular, $\Gamma_{d,\sigma}$ is stratified by G_σ-orbits indexed by cyclic rank patterns $r \in R_d^{\mathrm{orb}}$ satisfying $\mathrm{rank}(r) = \mathrm{rank}(\sigma)$.

We prove this theorem in the appendix 5.3

Appendix

Critical loci and cyclic conditions

We proove proposition 3.1

Proof. We proceed in explicit steps. Note that in this proof we have $\mathcal{X}_{r,\sigma} := \Gamma_{d,XY}$

Step 1: Proof of (22). By the assumed DLN normal form for rank-r critical points, $W \in C_r$ if and only if there exists $Z \in \mathcal{X}_{r,\sigma}$ and a sequence of hidden invertible changes of basis $g = (g_1, \dots, g_{L-1}) \in G_h$ such that the transformed weights satisfy the block gauge-fixed identities (17)–(18). Equivalently, $g^{-1} \cdot W = \iota(Z) \in S_{r,\sigma}$. Therefore $W = g \cdot \iota(Z) \in G_h \cdot S_{r,\sigma}$, proving $C_r \subseteq G_h \cdot S_{r,\sigma}$. Conversely, if $W = g \cdot \iota(Z)$ with $Z \in \mathcal{X}_{r,\sigma}$, then W has the required normal form up to the hidden reparametrisation g, and since Z satisfies both (14) and (15), W is critical. Thus $G_h \cdot S_{r,\sigma} \subseteq C_r$. This proves (22).

Step 2: Image of Φ equals C_r. By definition (23), $\Phi(g, Z) = g \cdot \iota(Z) \in G_h \cdot S_{r,\sigma} = C_r$ by Step 1. Hence $\Phi(G_h \times \mathcal{X}_{r,\sigma}) \subseteq C_r$. Conversely, Step 1 already shows every $W \in C_r$ can be written $W = g \cdot \iota(Z)$ with $Z \in \mathcal{X}_{r,\sigma}$. Therefore $\Phi(G_h \times \mathcal{X}_{r,\sigma}) = C_r$. **Step 3: $H_{r,\sigma}$-invariance of Φ.** Fix $(g, Z) \in G_h \times \mathcal{X}_{r,\sigma}$ and $h \in H_{r,\sigma}$. Then

$$\Phi\big((g, Z) \cdot h\big) = \Phi\big(gh,\ h^{-1} \cdot Z\big) = (gh) \cdot \iota(h^{-1} \cdot Z).$$

Using associativity of the group action (13), this equals $g \cdot \big(h \cdot \iota(h^{-1} \cdot Z)\big)$. Thus it suffices to show

$$h \cdot \iota(h^{-1} \cdot Z) = \iota(Z). \tag{$\star$}$$

Write each $h_\ell = \left(\begin{smallmatrix} I_r & 0 \\ 0 & K_\ell \end{smallmatrix}\right)$ (recall that the setwise stabiliser condition and the endpoint structure force $A = I_r$; see the paragraph after Definition 8.2).

Middle layers ($\ell = 2, \dots, L-1$): Here $(h^{-1} \cdot Z)_\ell = K_\ell^{-1} Z_\ell K_{\ell-1}$, and the computation is identical to Step 2 of Proposition 8.3:

$$h_\ell \begin{pmatrix} I_r & 0 \\ 0 & K_\ell^{-1} Z_\ell K_{\ell-1} \end{pmatrix} h_{\ell-1}^{-1} = \begin{pmatrix} I_r & 0 \\ 0 & Z_\ell \end{pmatrix} = \iota(Z)_\ell.$$

Last layer ($\ell = L$): Since $g_L = I$ in G_h, the action gives $(h \cdot \iota(h^{-1} \cdot Z))_L = \iota(h^{-1} \cdot Z)_L\, h_{L-1}^{-1}$. Now $(h^{-1} \cdot Z)_L = Z_L K_{L-1}$ (since $g_L = I$ implies the effective residual action at layer L is right multiplication by K_{L-1}), so

$$[U_S,\ U_Q Z_L K_{L-1}] \begin{pmatrix} I_r & 0 \\ 0 & K_{L-1}^{-1} \end{pmatrix} = [U_S,\ U_Q Z_L] = \iota(Z)_L.$$

First layer ($\ell = 1$): Since $g_0 = I$, the action gives $(h \cdot \iota(h^{-1} \cdot Z))_1 = h_1\, \iota(h^{-1} \cdot Z)_1$. Now $(h^{-1} \cdot Z)_1 = K_1^{-1} Z_1$ (since $g_0 = I$), so

$$\begin{pmatrix} I_r & 0 \\ 0 & K_1 \end{pmatrix} \begin{pmatrix} U_S^\top \Sigma_{YX} \Sigma_X^{-1} \\ K_1^{-1} Z_1 \end{pmatrix} = \begin{pmatrix} U_S^\top \Sigma_{YX} \Sigma_X^{-1} \\ Z_1 \end{pmatrix} = \iota(Z)_1.$$

This proves ($\star$), hence Φ is $H_{r,\sigma}$-invariant and descends to $\bar{\Phi}$.

Step 4: Injectivity on the principal slice locus. Assume $\Phi(g, Z) = \Phi(g', Z')$ with $Z, Z' \in X^{\circ}_{r,\sigma}$. Then $g \cdot \iota(Z) = g' \cdot \iota(Z')$, and setting $h := g^{-1}g' \in G_h$ we obtain

$$\iota(Z) = h \cdot \iota(Z'). \tag{$\star\star$}$$

In particular, h maps the slice point $\iota(Z')$ to another slice point $\iota(Z)$, so $h \in \mathrm{Trans}_{G_h}(\iota(Z'), S_{r,\sigma})$. Since $Z' \in \mathcal{X}^{\circ}_{r,\sigma}$, we have $\mathrm{Trans}_{G_h}(\iota(Z'), S_{r,\sigma}) = H_{r,\sigma}$ by definition of the principal slice locus. Therefore $h \in H_{r,\sigma}$.

Having established $h \in H_{r,\sigma}$, we compute ($\star\star$) componentwise. By the same block multiplication as in Step 3 (run in reverse), equation ($\star\star$) yields $Z = h \cdot Z'$, i.e. $Z' = h^{-1} \cdot Z$. Therefore $(g', Z') = (gh, h^{-1}\cdot Z) = (g, Z)\cdot h$ in the right action (24), and hence $[g', Z'] = [g, Z]$ in $G_h \times_{H_{r,\sigma}} \mathcal{X}^{\circ}_{r,\sigma}$. This proves injectivity on the principal slice locus.

Step 5: The principal slice locus is generically non-empty. We show that $\mathcal{X}^{\circ}_{r,\sigma}$ contains a dense open subset of $X_{r,\sigma}$, so that the bijection (25) holds on a large locus. Take any $h = (h_1, \dots, h_{L-1}) \in G_h$ satisfying $h \cdot \iota(Z') \in S_{r,\sigma}$ for some $Z' \in X_{r,\sigma}$. We must show $h \in H_{r,\sigma}$ for generic Z'.

Write $h_\ell = \left(\begin{smallmatrix} a_\ell & b_\ell \\ c_\ell & k_\ell \end{smallmatrix}\right)$ in the $r \times (d_\ell - r)$ block decomposition.

Endpoint $\ell = L$: The condition $(h \cdot \iota(Z'))_L = \iota(Z)_L$ gives $[U_S, U_Q Z'_L]\, h_{L-1}^{-1} = [U_S, U_Q Z_L]$. Write $h_{L-1}^{-1} = \left(\begin{smallmatrix} \alpha & \beta \\ \gamma & \kappa \end{smallmatrix}\right)$. Comparing the first block column and using $U_S^\top U_Q = 0$ yields $\alpha = I_r$ (from the U_S-component) and $Z'_L\gamma = 0$ (from the U_Q-component). For generic $Z' \in X_{r,\sigma}$, the matrix Z'_L has full row rank $d_L - r$, so $\gamma = 0$. Hence h_{L-1}^{-1} is block upper-triangular; since it is also invertible, its inverse h_{L-1} is block upper-triangular as well.

Endpoint $\ell = 1$: The condition $(h \cdot \iota(Z'))_1 = \iota(Z)_1$ gives $h_1 \left(\begin{smallmatrix} U_S^\top \Sigma_{YX}\Sigma_X^{-1} \\ Z'_1 \end{smallmatrix}\right) = \left(\begin{smallmatrix} U_S^\top \Sigma_{YX}\Sigma_X^{-1} \\ Z_1 \end{smallmatrix}\right)$. From the top block: $a_1 U_S^\top \Sigma_{YX}\Sigma_X^{-1} + b_1 Z'_1 = U_S^\top \Sigma_{YX}\Sigma_X^{-1}$. Generically Z'_1 is linearly independent from the rows of $U_S^\top \Sigma_{YX}\Sigma_X^{-1}$, which forces $b_1 = 0$ and $a_1 = I_r$. From the bottom block: $c_1 U_S^\top \Sigma_{YX}\Sigma_X^{-1} + k_1 Z'_1 = Z_1$; for this to admit a solution with Z_1 free, we need $c_1 = 0$ (since otherwise $c_1 U_S^\top \Sigma_{YX}\Sigma_X^{-1}$ would introduce a component in the signal subspace, contradicting the block structure). Hence h_1 is block-diagonal with $a_1 = I_r$.

Middle layers ($\ell = 2, \dots, L{-}2$): Having established that h_1 and h_{L-1} are block-diagonal, one propagates inward. At each layer the condition

$$h_\ell \begin{pmatrix} I_r & 0 \\ 0 & Z'_\ell \end{pmatrix} h_{\ell-1}^{-1} = \begin{pmatrix} I_r & 0 \\ 0 & Z_\ell \end{pmatrix}$$

with $h_{\ell-1}$ already known to be block-diagonal forces $b_\ell Z'_\ell \kappa_{\ell-1} = 0$ and $c_\ell = 0$. Since $\kappa_{\ell-1}$ is invertible and generically Z'_ℓ has full row rank, we get $b_\ell = 0$. Hence h_ℓ is block-diagonal at every layer.

Thus, for Z' in a dense open subset of $X_{r,\sigma}$ (where certain residual blocks have full rank), every element of $\mathrm{Trans}_{G_h}(\iota(Z'), S_{r,\sigma})$ is block-diagonal, i.e. belongs to $H_{r,\sigma}$. Since $H_{r,\sigma} \subseteq \mathrm{Trans}_{G_h}(\iota(Z'), S_{r,\sigma})$ always holds (by Step 3), we conclude $\mathrm{Trans}_{G_h}(\iota(Z'), S_{r,\sigma}) = H_{r,\sigma}$ on this locus, confirming that $X^{\circ}_{r,\sigma}$ is generically non-empty.

Step 6: Conclusion. Steps 2 and 3 establish that $\bar{\Phi}$ is a well-defined surjection. Step 4 establishes injectivity on $\mathcal{X}^{\circ}_{r,\sigma}$, giving (25). Taking G_h-quotients gives (26). The global map $\bar{\Phi}\colon G_h \times_{H_{r,\sigma}} \mathcal{X}_{r,\sigma} \to C_r$ is surjective but may fail to be injective at points Z' where certain residual blocks Z'_ℓ are rank-deficient, causing $\mathrm{Trans}_{G_h}(\iota(Z'), S_{r,\sigma})$ to strictly contain $H_{r,\sigma}$. $\square$

DLNs as Quiver Representations

Definition 5.22. A *quiver* is a directed graph $Q = (P, E, s, t)$ where P is the set of vertices, E the set of directed edges, and $s, t : E \to P$ are the source and target maps.

Definition 5.23. A *representation* of a quiver Q over a field k assigns a vector space V_v to each vertex $v \in P$ and a linear map $\phi_e : V_{s(e)} \to V_{t(e)}$ to each directed edge $e \in E$. For a fixed *dimension vector* $d = (d_v)_{v \in P}$ with $d_v = \dim_k(V_v)$, we denote by $\mathrm{Rep}_d(Q)$ the space of d-dimensional representations of Q over k:

$$\mathrm{Rep}_d(Q) \cong \bigoplus_{e \in E} \mathrm{Hom}_k(k^{d_{s(e)}}, k^{d_{t(e)}}).$$

Definition 5.24. The *path algebra* kQ is the k-vector space with basis given by all paths in Q (including a trivial path e_v of length 0 at each vertex v). Multiplication is given by concatenation of paths when composable and 0 otherwise, extended by linearity.

Proposition 5.25. The category of representations of a quiver Q is equivalent to the category of left modules over its path algebra kQ.

Sketch of proof: Given a representation (V_v, ϕ_e), define the module $M = \bigoplus_{v \in P} V_v$ with e_v acting as projection onto V_v and an arrow $a : v \to w$ acting as ϕ_a on V_v and 0 elsewhere. Conversely, for a kQ-module M, define $V_v = e_v M$ and ϕ_a the action of a on M. These constructions define functors that are quasi-inverse to each other.

Definition 5.26. A *deep linear network (DLN)* of depth L is a linear map $\phi \in \mathrm{Hom}_k(k^{d_0}, k^{d_L})$ that factorizes as a composition of linear maps

$$\phi = \phi_L \phi_{L-1} \cdots \phi_1,$$

where $\phi_\ell \in \mathrm{Hom}_k(k^{d_{\ell-1}}, k^{d_\ell})$ for $\ell = 1, \ldots, L$.

Proposition 5.27. A DLN with dimensions $(d_0, \ldots, d_L)$ can be identified with a representation of the *linear quiver* A_{L+1} consisting of $L+1$ vertices connected by L consecutive arrows. In this identification, each layer ϕ_ℓ corresponds to an edge of A_{L+1} and the vector spaces k^{d_ℓ} to the vertices.

Over $k = \mathbb{R}$ (or $\mathbb{C}$) we fix inner products on each $V_v \cong k^{d_v}$ and endow each $\mathrm{Hom}(V_{s(e)}, V_{t(e)})$ with the Frobenius inner product

$$\langle \phi, \psi \rangle = \mathrm{Tr}(\phi^\top \psi).$$

Summing this over all edges defines an inner product on $\mathrm{Rep}_d(Q)$ and hence a Euclidean structure on the representation space.

Definition 5.28. Let $\phi^\star \in \mathrm{Hom}_k(k^{d_0}, k^{d_L})$ be a target linear map. The loss function of a DLN $\phi = \phi_L \cdots \phi_1$ is

$$\mathcal{L}((\phi_\ell)_{\ell=1}^L) = \|\phi^\star - \phi_L \cdots \phi_1\|_F^2,$$

where $\|\cdot\|_F$ denotes the Frobenius norm. We are particularly interested in the fiber $\mathcal{L}^{-1}(0)$.

Proposition 5.29. The representation space $\mathrm{Rep}_d(Q)$ is an affine variety with coordinate ring

$$k[\mathrm{Rep}_d(Q)] = k[x_{ij}^{(\ell)} \mid 1 \le \ell \le L,\ 1 \le i \le d_\ell,\ 1 \le j \le d_{\ell-1}],$$

and hence

$$\mathrm{Rep}_d(Q) \cong \mathbb{A}^{\sum_{\ell=1}^L d_{\ell-1} d_\ell}.$$

Proposition 5.30. The loss $\mathcal{L} : \mathrm{Rep}_d(Q) \to k$ is a polynomial map of degree $2L$. Its graph

$$\{(\Phi, \lambda) \in \mathrm{Rep}_d(Q) \times k \mid \lambda = \mathcal{L}(\Phi)\}$$

is an affine subvariety of dimension $\sum_{\ell=1}^{L} d_{\ell-1} d_\ell$.

Since $\mathrm{Rep}_d(Q)$ is isomorphic to an affine space, it carries a natural module of Kähler differentials $\Omega_{k[\mathrm{Rep}_d]/k}$, giving an algebraic notion of differential $d\mathcal{L}$. Over $\mathbb{R}$ or $\mathbb{C}$, once the Frobenius inner product is fixed, this identifies with the usual analytic gradient and the second differential with the Hessian of $\mathcal{L}$. This will enable us to study the critical points and local geometry of the DLN loss landscape.

Fibers of the multiplication map

This is a summary of Simon Pepin Lehalleur's paper on computing the codimension of the global minimum of the population loss of a deep linear network. The weight matrices of a DLN defines a representation of a Quiver. We have an action of the group $GL_{\bar{d}} := \prod_i GL_{d_i}$ over $Rep_{\bar{d}}$. We can then partition $Rep_{\bar{d}}$ into its orbits under the action of $G_d := GL_{\bar{d}}$. Recall that we have an isomorphism between Rep_d and kQ-modules. The isomorphism classes of kQ-modules of dimension $\bar{d}$ are isomorphic to the orbits of Rep_d under the action of G_d. The loci of zero loss is defined by:

$$\pi^{-1}(0) \subset Rep_{\bar{d}} \text{ with } \pi(\phi) = \phi_L ... \phi_1$$

We are interested by relating the geometry of $\pi^{-1}(0)$ with the combinatorics of the ranks of products of weight matrices. Note that we have:

$$\pi^{-1}(0) = \Sigma_d^0 = \{\phi \in Rep_d | rk(\phi) = 0\}$$

More generally, define:

$$\Sigma_d^r = \{\phi \in Rep_d | rk(\phi) = r\}$$

Given that the map π is equivarient w.r.t to G, we can decompose Σ_d^r into orbits under the action of G. This is true because we have a group action of G induced on $\mathrm{Im}(\pi)$ by the equivarience relation. An important result is that we can parametrize the G-orbits of Rep_d via the multiplicity pattern defined by:

$$\mathcal{M}_d^+ := \{(m_{ij})_{1 \leq i \leq j \leq N} | m_{ij} \in \mathbb{N}, d_k = \sum_{i \leq k \leq j} m_{ij}\}$$

The multiplicity patterns parametrize the isomorphic classes of the of the Dynkin Quiver $\mathbb{A}_n$-modules whose indecomposables representaions are given by (following from Gabriel's Theorem):

$$M_{ij} := 0 \to ... \to 0 \to k \to ... \to k \to 0 ... \to 0$$

By Krull-Schmidt every Q-module writes:

$$M = \oplus_{1 \leq i \leq j \leq N} m_{ij} M_{ij}$$

The rank patterns is another important object that enables to parametrize the orbits of Rep_d as well in the particular case of the A_n quiver. More specifically we are interested in the following orbit:

$$\mathcal{O}_{\bar{r}} := \{\phi | rk(\phi_j ... \phi_{i+1} = r_{ij}\}$$

With:

$$r_{ij} \in \mathcal{R}_d := \{(r_{ij})_{1\leq i\leq j\leq N} | r_{ij} \in \mathbb{N}, r_{ij} - r_{i,j+1} - r_{i-1,j} + r_{i-1,j+1} \geq 0, r_{ii} = d_i\}$$

The rank pattern set $\mathcal{R}_d$ is in bijection with the multiplicity patterns and therefore parametrize the orbit of Rep_d as well. In other words, the G-orbits of Rep_d corresponds to rank condition on a composition of weight matrices in a DLN. We can therefore stratify the space Σ_d^r by the orbits $\mathcal{O}_{\bar{r}}$ parametrized by the rank patterns. This allows to relate the codimension of 0-loss with the codimension of the orbits which can be encoded in a power series using Pochammer symbols.

Parameterization of saddle points

Recall proposition 9 and 10 from the paper The Loss Landscape of Deep Linear Neural Networks: a Second-order Analysis.
Assumptions:

- Input data dimension d_x and output data dimension d_y are such that $d_y \leq d_x \leq m$
- Input data covariance Σ_X is invertible and input-output data covariance Σ_{YX} is full rank d_y
- The singular values of $\sigma = \Sigma_{YX}\Sigma_X^{-1}X$ are all distinct

Proposition 9 (reformulated): Let ϕ be a critical point. Let $S \subset \{1, ...d_y\}$ be a set indexing the singular values of the target map. Let $r := |S|$ be the rank of ϕ, proposition 9 ensures that ϕ is conjugate to the point $\tilde{\phi}$ in Rep_d with:

$$\begin{aligned}
\pi(\tilde{\phi}) &= \tilde{\phi}_L ... \tilde{\phi}_1 = U_S U_S^\top \Sigma_{YX}\Sigma_X^{-1}, \quad \text{the map } U_S U_S^\top \text{ projects on singular values indexed by } S \\
\tilde{\phi}_L &= [U_S \ U_Q Z_L] \quad \exists Z_L \in \mathbb{R}^{d_y - r \times d_{L-1} - r} \text{ and } Q = \{1...d_y\} - S \\
\forall h \in \{L-1, ..., 2\},\ \tilde{\phi}_h &= I_r \oplus Z_h, \quad \exists Z_h \in \mathbb{R}^{d_h - r \times d_{h-1} - r} \\
\tilde{\phi}_1 &= [U_S^\top \Sigma_{YX}\Sigma_X^{-1} \ Z_1]^\top \exists Z_1 \in \mathbb{R}^{d_1 - r \times d_0 - r} \\
U_Q Z_L ... Z_2 &= 0
\end{aligned}$$

Let $\tilde{\phi_Z}$ be a representation Q satisfying the conditions of the proposition just above. Define:

$$U^S_{\tilde{\phi}_Z} = \{\phi \in Rep_d | \phi \sim \tilde{\phi_Z}\}$$

In particular, we see that the set of critical point of rank $r = |S|$ associated with the subset of singular values indexed by S satisfies:

$$\mathcal{C}_{L,S} \subset \cup_Z U^S_{\tilde{\phi}_Z} \subset \pi^{-1}(U_S U_S^\top \Sigma_{XY}\Sigma_X^{-1})$$

Proposition 10 (reformulated): Consider ϕ such that ϕ is conjugate to the representation $\tilde{\phi}$ such that:

$$\begin{aligned}
\tilde{\phi}_L &= [U_S \ U_Q Z_L] \quad \exists Z_L \in \mathbb{R}^{d_y - r \times d_{L-1} - r} \text{ and } Q \in \{1...d_y\} - S \\
\forall h \in \{L-1, ..., 2\},\ \tilde{\phi}_h &= I_r \oplus Z_h, \quad \exists Z_h \in \mathbb{R}^{d_h - r \times d_{h-1} - r} \\
\tilde{\phi}_1 &= [U_S^\top \Sigma_{XY}\Sigma_X^{-1} \ Z_1]^\top \exists Z_1 \in \mathbb{R}^{d_1 - r \times d_0 - r}
\end{aligned}$$

if $r = r_{max}$ or if there exists $h_1 \neq h_2$ with $Z_{h_1} = Z_{h_2} = 0$ then ϕ is a first order critical point of the loss associated with the subset S.
Let ϕ be a representation Q satisfying the conditions of the proposition just above. Define:

$$U^S_{h_1,h_2} = \{\phi \in Rep_d | \phi \sim \tilde{\phi} \text{ and } \exists h_1 \neq h_2 | Z_{h_1} = Z_{h_2} = 0\}$$

Then we have:

$$U^S_{h_1,h_2} \subset \mathcal{C}_{L,S} \subset \cup_Z U^S_{\tilde{\phi}_Z} \subset \pi^{-1}(U_S U_S^\top \Sigma_{XY} \Sigma_X^{-1})$$

Unfortunately, a full parameterization of the first order critical points is lacking. This is because we could find a critical point that does not fall into $U^S_{h_1,h_2}$ for example or some representation in $U^S_{\tilde{\phi}}$ that is not a critical point.
Recall that a critical point satisfies:

$$\pi_{>l}(\phi)^t E(\phi) \pi_{<l}(\phi)^t = 0$$

Let $\phi^\star := \Sigma_{XY}\Sigma_X^{-1}$ be the target map and let $P_S := U_S U_S^\top$ be a projector on singular values indexed by S. Using proposition 9 we can write this equation as:

$$\pi_{>l}(\phi)^t (I - P_S) \phi^\star \pi_{<l}(\phi)^t = 0$$

Let $P_S^\perp := 1 - P_S$, unfortunately an GL_d group action on the map:

$$\psi_S(\phi) := \pi_{>l}(\phi)^t P_S^\perp \phi^\star \pi_{<l}(\phi)^t$$

allows to define a group of GL_d over the critical points if only if

$$\pi_{>l}(\phi)^t g_L^{-t} P_S^\perp \phi^\star g_0^{-t} \pi_{<l}(\phi)^t = 0$$

Which does not seem to hold in general.
Proposition The gradient is equivariant under the action of G_h and critical points are G_h invariant.
Proof We have already shown that the gradient map satisfies:

$$g * \nabla_l L(\phi) = g_l^t \nabla_l L(\phi) g_{l-1}^t = \nabla_l L(g.\phi)$$

Let ϕ_S be a critical point i.e. we have

$$\nabla L(\phi_S) = 0$$

Acting on ϕ_S via the hidden subgroup G_h yields:

$$\nabla_l L(g\phi_S) = g_l^t \nabla_l L(\phi_S) g_{l-1}^t = 0$$

Since $\nabla_l L(\phi_S) = 0$ for all l.

Stratification of vanishing cyclic condition with fixed target map

We prove lemma 5.3:

Proof. We proceed in several steps.

Step 1: $\Gamma_{d,\sigma}$ is stable under the action of G_σ.
Let $x \in \Gamma_{d,\sigma}$, so by definition $F(x) = \sigma$. Let $g \in G_\sigma$. Using the equivariance property (**??**),

$$F(g \cdot x) \;=\; g_0\, F(x)\, g_L^{-1} \;=\; g_0\, \sigma\, g_L^{-1}.$$

Since $g \in G_\sigma$, we have $g_0\, \sigma\, g_L^{-1} = \sigma$, and hence $F(g \cdot x) = \sigma$. Therefore $g \cdot x \in \Gamma_{d,\sigma}$. This shows that $\Gamma_{d,\sigma}$ is invariant under the action of G_σ and, in particular,

$$G_\sigma \cdot x \subseteq \Gamma_{d,\sigma} \qquad \text{for all } x \in \Gamma_{d,\sigma}. \tag{$*$}$$

Step 2: If $x \in \mathcal{O} \cap \Gamma_{d,\sigma}$, then $G_\sigma \cdot x \subseteq \mathcal{O} \cap \Gamma_{d,\sigma}$.
Let $x \in \mathcal{O} \cap \Gamma_{d,\sigma}$. Since $\mathcal{O}$ is a G_d–orbit, it is stable under the action of G_d and hence under the subgroup $G_\sigma \subseteq G_d$. Thus $G_\sigma \cdot x \subseteq \mathcal{O}$. By $(*)$, $G_\sigma \cdot x \subseteq \Gamma_{d,\sigma}$. Combining these two inclusions gives

$$G_\sigma \cdot x \subseteq \mathcal{O} \cap \Gamma_{d,\sigma}. \tag{$**$}$$

Step 3: Any two points of $\mathcal{O} \cap \Gamma_{d,\sigma}$ differ by the action of G_σ.
Assume $x, y \in \mathcal{O} \cap \Gamma_{d,\sigma}$. Because x and y lie in the same G_d–orbit $\mathcal{O}$, there exists $g = (g_0, \dots, g_L) \in G_d$ such that

$$y = g \cdot x.$$

Applying F and using (**??**), we obtain

$$F(y) \;=\; F(g \cdot x) \;=\; g_0\, F(x)\, g_L^{-1}.$$

Since $x, y \in \Gamma_{d,\sigma}$, we have $F(x) = F(y) = \sigma$. Therefore

$$\sigma = g_0\, \sigma\, g_L^{-1}.$$

By definition of G_σ, this implies $g \in G_\sigma$. Hence $y = g \cdot x \in G_\sigma \cdot x$, and we conclude that

$$\mathcal{O} \cap \Gamma_{d,\sigma} \subseteq G_\sigma \cdot x. \tag{$*\!*\!*$}$$

Step 4: Conclusion.
If $\mathcal{O} \cap \Gamma_{d,\sigma} = \varnothing$, there is nothing to prove. Otherwise choose any $x \in \mathcal{O} \cap \Gamma_{d,\sigma}$. Inclusion $(**)$ gives $G_\sigma \cdot x \subseteq \mathcal{O} \cap \Gamma_{d,\sigma}$, while inclusion $(*\,*\,*)$ gives the reverse inclusion. Therefore

$$\mathcal{O} \cap \Gamma_{d,\sigma} = G_\sigma \cdot x.$$

This proves that whenever the intersection is not empty, it consists of a single G_σ–orbit. The lemma follows. $\square$

Stratification of vanishing cyclic condition with fixed target map

we prove proposition 5.4

Proof. We prove (i) and (ii) in separate steps.

Step 1: A basic invariance identity for ranks under left–right change of bases. Let $A \in GL(V_0)$, $B \in GL(V_L)$, and $T \in W = \mathrm{Hom}_{\mathbb{R}}(V_L, V_0)$. Consider the map $T' := A\,T\,B^{-1} \in W$. We have $\mathrm{rank}(T') = \mathrm{rank}(T)$.

Step 2: Proof of (i) (well-definedness of $r(\lambda)$). Fix $\lambda \in \Lambda$ and choose two points $x, y \in \mathcal{O}_\lambda$. By definition of orbit, there exists $g = (g_0, \ldots, g_L) \in G_d$ such that

$$y = g \cdot x.$$

Apply F to both sides and use the equivariance identity (7):

$$F(y) = F(g \cdot x) = g_0\,F(x)\,g_L^{-1}.$$

Now apply Step 1 with $A = g_0$, $B = g_L$, and $T = F(x)$. This yields

$$\mathrm{rank}(F(y)) = \mathrm{rank}(g_0\,F(x)\,g_L^{-1}) = \mathrm{rank}(F(x)).$$

Thus the rank of $F(\,\cdot\,)$ is constant along $\mathcal{O}_\lambda$, so the integer $r(\lambda)$ defined by (8) is independent of the choice of $x \in \mathcal{O}_\lambda$. This proves (i).

Step 3: A real classification fact for linear maps by rank. Let $S, T \in W = \mathrm{Hom}_{\mathbb{R}}(V_L, V_1)$. We use the following statement:

Claim. If $\mathrm{rank}(S) = \mathrm{rank}(T)$, then there exist $A \in GL(V_0)$ and $B \in GL(V_L)$ such that

$$S = A\,T\,B^{-1}. \tag{17}$$

Step 4: Proof of (ii), forward implication. Assume $\mathcal{O}_\lambda \cap \Gamma_\sigma \neq \varnothing$. Choose $x \in \mathcal{O}_\lambda \cap \Gamma_\sigma$. By definition of $\Gamma_\sigma = F^{-1}(\sigma)$ we have $F(x) = \sigma$, hence

$$\mathrm{rank}(\sigma) = \mathrm{rank}(F(x)).$$

By definition of $r(\lambda)$ (and Step 2), $\mathrm{rank}(F(x)) = r(\lambda)$ for any $x \in \mathcal{O}_\lambda$. Therefore $\mathrm{rank}(\sigma) = r(\lambda)$, proving the forward direction of (9).

Step 5: Proof of (ii), reverse implication. Assume $r(\lambda) = \mathrm{rank}(\sigma)$. Choose any $x_0 \in \mathcal{O}_\lambda$. Then, by definition of $r(\lambda)$,

$$\mathrm{rank}(F(x_0)) = r(\lambda) = \mathrm{rank}(\sigma).$$

By the claim in Step 3 (applied to $T = F(x_0)$ and $S = \sigma$), there exist $A \in GL(V_0)$ and $B \in GL(V_L)$ such that

$$\sigma = A\,F(x_0)\,B^{-1}.$$

Define $g = (g_0, \ldots, g_L) \in G_d$ by setting

$$g_0 := A, \qquad g_L := B, \qquad g_i := \mathrm{Id}_{V_i} \ \text{ for } i = 1, \ldots, L-1.$$

Then $g \in G_d$ and, using equivariance (7),

$$F(g \cdot x_0) = g_0\,F(x_0)\,g_L^{-1} = A\,F(x_0)\,B^{-1} = \sigma.$$

Hence $g \cdot x_0 \in \Gamma_\sigma$. Also, since $x_0 \in \mathcal{O}_\lambda$ and $\mathcal{O}_\lambda$ is a G_d–orbit, we have $g \cdot x_0 \in \mathcal{O}_\lambda$. Therefore $g \cdot x_0 \in \mathcal{O}_\lambda \cap \Gamma_\sigma$, so $\mathcal{O}_\lambda \cap \Gamma_\sigma \neq \varnothing$.

Combining Steps 4 and 5 yields (9). This completes the proof. □

Rank formula for the stratification with target map

We prove proposition 5.7:

Proof. Fix $\mathbf{m} = (m_{i,\ell}) \in \mathcal{M}_d^+$. Choose any $\phi \in \mathcal{O}_{\mathbf{m}}$ and write $M := M(\phi)$ for the corresponding Nakayama module. By definition of $\mathcal{O}_{\mathbf{m}}$, there exists an isomorphism of modules

$$M \;\cong\; \bigoplus_{i,\ell} M(i,\ell)^{\oplus m_{i,\ell}}. \tag{18}$$

We will compute $\operatorname{rank}(\phi_0) = \operatorname{rank}(F(\phi))$ from this decomposition.

Step 1: Describe the vector spaces at vertices 0 and L using (18).

For each vertex $j \in \{0, L\}$ the representation space is $V_j = M(j)$. Evaluating (18) at the vertex j gives an isomorphism of real vector spaces

$$V_j \;=\; M(j) \;\cong\; \bigoplus_{i,\ell} M(i,\ell)(j)^{\oplus m_{i,\ell}}. \tag{19}$$

For cyclic Nakayama indecomposables, $M(i,\ell)(j)$ is either 0 or a 1–dimensional real vector space (depending on whether $j \in \operatorname{Supp}(i,\ell)$). Therefore each summand in (19) is either 0 or $\mathbb{R}^{m_{i,\ell}}$.

Step 2: The closing arrow respects the direct-sum decomposition.

The linear map $\phi_0 : V_L \to V_0$ is the structure map on the quiver arrow $L \to 0$. Because M is a direct sum of submodules as in (18), the action of every quiver arrow, and in particular the arrow $L \to 0$, decomposes as the direct sum of its actions on each direct summand. Concretely, after choosing identifications as in (19) for $j = L$ and $j = 0$, the map ϕ_0 can be written as a block-diagonal direct sum

$$\phi_0 \;\cong\; \bigoplus_{i,\ell} \left(\phi_0\right)_{i,\ell} \tag{20}$$

where

$$\left(\phi_0\right)_{i,\ell} :\; M(i,\ell)(L)^{\oplus m_{i,\ell}} \longrightarrow M(i,\ell)(0)^{\oplus m_{i,\ell}}$$

is the map induced by $L \to 0$ on the summand $M(i,\ell)^{\oplus m_{i,\ell}}$.

Step 3: Compute each block $\left(\phi_0\right)_{i,\ell}$.

Fix a pair (i,ℓ). There are two cases.

Case A: $\epsilon(i,\ell) = 0$. By definition, $\epsilon(i,\ell) = 0$ means the arrow $L \to 0$ acts trivially on $M(i,\ell)$, i.e. the map

$$M(i,\ell)(L) \to M(i,\ell)(0)$$

is the zero map. Taking $m_{i,\ell}$ copies, the induced map

$$\left(\phi_0\right)_{i,\ell} :\; M(i,\ell)(L)^{\oplus m_{i,\ell}} \to M(i,\ell)(0)^{\oplus m_{i,\ell}}$$

is also the zero map. Hence

$$\operatorname{rank}\left(\left(\phi_0\right)_{i,\ell}\right) = 0. \tag{21}$$

Case B: $\epsilon(i,\ell) = 1$. By definition, $\epsilon(i,\ell) = 1$ means the arrow $L \to 0$ acts nontrivially on $M(i,\ell)$. For Nakayama indecomposables, both spaces $M(i,\ell)(L)$ and $M(i,\ell)(0)$ are then 1–dimensional and the map between them is a nonzero linear map $\mathbb{R} \to \mathbb{R}$. Choose bases of these 1–dimensional spaces; in these bases the map is multiplication by a nonzero scalar $a_{i,\ell} \in \mathbb{R}^\times$. On $m_{i,\ell}$ copies, the induced map is multiplication by the same scalar on each copy; in coordinates it is

$$a_{i,\ell} \cdot I_{m_{i,\ell}} : \ \mathbb{R}^{m_{i,\ell}} \to \mathbb{R}^{m_{i,\ell}},$$

which is invertible because $a_{i,\ell} \neq 0$. Therefore

$$\operatorname{rank}\left(\left(\phi_0\right)_{i,\ell}\right) = m_{i,\ell}. \tag{22}$$

Step 4: Sum the ranks of the blocks.

By (20), after choosing bases compatible with the decomposition, ϕ_0 is block diagonal with blocks $\left(\phi_0\right)_{i,\ell}$. The rank of a block-diagonal linear map is the sum of the ranks of its diagonal blocks (because its image is the direct sum of the images of the blocks). Hence

$$\operatorname{rank}(\phi_0) \ = \ \sum_{i,\ell} \operatorname{rank}\left(\left(\phi_0\right)_{i,\ell}\right).$$

Substituting (21) in Case A and (22) in Case B yields

$$\operatorname{rank}(\phi_{L+1}) \ = \ \sum_{i,\ell} \epsilon(i,\ell)\, m_{i,\ell}.$$

By definition $r(\mathbf{m}) = \operatorname{rank}(\phi_{L+1})$ for $\phi \in \mathcal{O}_{\mathbf{m}}$, so this proves (10). $\square$

Proof of the inversion formula

Proof of 5.11. Fix a pair (i,j) with $\delta(i,j) \leq L$. By the cumulative rank formula,

$$r_{ab} = \sum_{\substack{k,\ell \\ [a \rightsquigarrow b] \subseteq [k \rightsquigarrow \ell]}} m_{k\ell} \tag{23}$$

for every pair (a,b) with $\delta(a,b) \leq L$.

We apply this formula to the four pairs

$$(a,b) = (i,j), \qquad (i-1,j), \qquad (i,j+1), \qquad (i-1,j+1).$$

Thus

$$r_{ij} = \sum_{\substack{k,\ell \\ [i \rightsquigarrow j] \subseteq [k \rightsquigarrow \ell]}} m_{k\ell}, \tag{24}$$

$$r_{i-1,j} = \sum_{\substack{k,\ell \\ [i-1 \rightsquigarrow j] \subseteq [k \rightsquigarrow \ell]}} m_{k\ell}, \tag{25}$$

$$r_{i,j+1} = \sum_{\substack{k,\ell \\ [i \rightsquigarrow j+1] \subseteq [k \rightsquigarrow \ell]}} m_{k\ell}, \tag{26}$$

$$r_{i-1,j+1} = \sum_{\substack{k,\ell \\ [i-1 \rightsquigarrow j+1] \subseteq [k \rightsquigarrow \ell]}} m_{k\ell}. \tag{27}$$

Consider now the linear combination

$$X := r_{ij} - r_{i-1,j} - r_{i,j+1} + r_{i-1,j+1}.$$

Substituting (24)–(27), we obtain

$$X = \sum_{k,\ell} c_{k\ell}\, m_{k\ell},$$

where the coefficient $c_{k\ell}$ is given by

$$c_{k\ell} = \mathbf{1}_{[i\rightsquigarrow j]\subseteq[k\rightsquigarrow \ell]} - \mathbf{1}_{[i-1\rightsquigarrow j]\subseteq[k\rightsquigarrow \ell]} - \mathbf{1}_{[i\rightsquigarrow j+1]\subseteq[k\rightsquigarrow \ell]} + \mathbf{1}_{[i-1\rightsquigarrow j+1]\subseteq[k\rightsquigarrow \ell]}.$$

Here $\mathbf{1}_P$ denotes the indicator of the statement P.

We will show that

$$c_{k\ell} = \begin{cases} 1, & \text{if } (k,\ell) = (i,j), \\ 0, & \text{otherwise.} \end{cases}$$

It will then follow immediately that

$$X = m_{ij},$$

which is exactly (13).

We therefore fix an interval $[k \rightsquigarrow \ell]$ and analyze its contribution case by case.

Step 1: if $[i \rightsquigarrow j] \not\subseteq [k \rightsquigarrow \ell]$, then $c_{k\ell} = 0$.

Indeed, if $[i \rightsquigarrow j]$ is not contained in $[k \rightsquigarrow \ell]$, then neither $[i-1 \rightsquigarrow j]$, nor $[i \rightsquigarrow j+1]$, nor $[i-1 \rightsquigarrow j+1]$ can be contained in $[k \rightsquigarrow \ell]$, because each of these three intervals contains $[i \rightsquigarrow j]$ as a subinterval. Hence all four indicator functions vanish, so $c_{k\ell} = 0$.

Step 2: suppose $[i \rightsquigarrow j] \subseteq [k \rightsquigarrow \ell]$.

Then the first indicator is equal to 1. We must determine when the other three indicators are equal to 1.

Observe:

- The inclusion

  $$[i-1 \rightsquigarrow j] \subseteq [k \rightsquigarrow \ell]$$

  holds if and only if $[k \rightsquigarrow \ell]$ extends at least one step to the left of $[i \rightsquigarrow j]$, that is, if and only if $k \neq i$. Equivalently, it fails precisely when $k = i$.

- The inclusion

  $$[i \rightsquigarrow j+1] \subseteq [k \rightsquigarrow \ell]$$

  holds if and only if $[k \rightsquigarrow \ell]$ extends at least one step to the right of $[i \rightsquigarrow j]$, that is, if and only if $\ell \neq j$. Equivalently, it fails precisely when $\ell = j$.

- The inclusion

  $$[i-1 \rightsquigarrow j+1] \subseteq [k \rightsquigarrow \ell]$$

  holds if and only if $[k \rightsquigarrow \ell]$ extends both one step to the left and one step to the right of $[i \rightsquigarrow j]$, that is, if and only if $k \neq i$ and $\ell \neq j$. Equivalently, it holds exactly when both of the previous two inclusions hold.

These statements are immediate from the geometry of cyclic intervals: once $[i \rightsquigarrow j] \subseteq [k \rightsquigarrow \ell]$, enlarging the left endpoint from i to $i-1$ asks whether the containing interval starts strictly before i, and enlarging the right endpoint from j to $j+1$ asks whether the containing interval ends strictly after j.

Thus there are four possibilities.

Case 2a: $k = i$ and $\ell = j$.

Then $[k \rightsquigarrow \ell] = [i \rightsquigarrow j]$. Hence

$$[i \rightsquigarrow j] \subseteq [k \rightsquigarrow \ell]$$

is true, while

$$[i-1 \rightsquigarrow j] \subseteq [k \rightsquigarrow \ell], \qquad [i \rightsquigarrow j+1] \subseteq [k \rightsquigarrow \ell], \qquad [i-1 \rightsquigarrow j+1] \subseteq [k \rightsquigarrow \ell]$$

are all false. Therefore

$$c_{k\ell} = 1 - 0 - 0 + 0 = 1.$$

Case 2b: $k \neq i$ and $\ell = j$.

Then $[k \rightsquigarrow \ell]$ extends to the left of $[i \rightsquigarrow j]$ but not to the right. Hence

$$[i \rightsquigarrow j] \subseteq [k \rightsquigarrow \ell], \qquad [i-1 \rightsquigarrow j] \subseteq [k \rightsquigarrow \ell]$$

are true, while

$$[i \rightsquigarrow j+1] \subseteq [k \rightsquigarrow \ell], \qquad [i-1 \rightsquigarrow j+1] \subseteq [k \rightsquigarrow \ell]$$

are false. Therefore

$$c_{k\ell} = 1 - 1 - 0 + 0 = 0.$$

Case 2c: $k = i$ and $\ell \neq j$.

Then $[k \rightsquigarrow \ell]$ extends to the right of $[i \rightsquigarrow j]$ but not to the left. Hence

$$[i \rightsquigarrow j] \subseteq [k \rightsquigarrow \ell], \qquad [i \rightsquigarrow j+1] \subseteq [k \rightsquigarrow \ell]$$

are true, while

$$[i-1 \rightsquigarrow j] \subseteq [k \rightsquigarrow \ell], \qquad [i-1 \rightsquigarrow j+1] \subseteq [k \rightsquigarrow \ell]$$

are false. Therefore

$$c_{k\ell} = 1 - 0 - 1 + 0 = 0.$$

Case 2d: $k \neq i$ and $\ell \neq j$.

Then $[k \rightsquigarrow \ell]$ extends both to the left and to the right of $[i \rightsquigarrow j]$. Hence all four inclusions hold:

$$[i \rightsquigarrow j] \subseteq [k \rightsquigarrow \ell], \quad [i-1 \rightsquigarrow j] \subseteq [k \rightsquigarrow \ell], \quad [i \rightsquigarrow j+1] \subseteq [k \rightsquigarrow \ell], \quad [i-1 \rightsquigarrow j+1] \subseteq [k \rightsquigarrow \ell].$$

Therefore

$$c_{k\ell} = 1 - 1 - 1 + 1 = 0.$$

Combining Step 1 and Step 2, we have proved that the coefficient of $m_{k\ell}$ in X is equal to 1 when $(k, \ell) = (i, j)$ and equal to 0 for every other pair (k, ℓ). Hence

$$X = m_{ij}.$$

Since

$$X = r_{ij} - r_{i-1,j} - r_{i,j+1} + r_{i-1,j+1},$$

this gives exactly

$$m_{ij} = r_{ij} - r_{i-1,\,j} - r_{i,\,j+1} + r_{i-1,\,j+1}.$$

This proves the formula. $\square$

Proof of the forward forumla for rank vs multiplicities

This is the proof of lemma 5.14:

Proof. Let $\phi \in \Gamma_{d,L+1}$ be any representation whose associated A-module has Krull–Schmidt multiplicity pattern m. By definition of m, the corresponding module admits a decomposition

$$M(\phi) \cong \bigoplus_{\substack{a,b \\ \delta(a,b) \le L-1}} M(a,b)^{\oplus m_{ab}}.$$

For fixed i, j with $\delta(i,j) \le L-1$, consider the path map

$$\Phi_{i \to j}(\phi) : V_i \to V_j$$

given by composition along the oriented segment $[i \rightsquigarrow j]$. Because $M(\phi)$ is a direct sum of the indecomposables $M(a,b)$, and because each arrow map of ϕ is block-diagonal with respect to this decomposition, the composed map $\Phi_{i \to j}(\phi)$ is also block-diagonal with respect to the same decomposition. Therefore

$$\operatorname{rank}\bigl(\Phi_{i \to j}(\phi)\bigr) = \sum_{\substack{a,b \\ \delta(a,b) \le L}} m_{ab}\, \operatorname{rank}\bigl(\Phi_{i \to j}|_{M(a,b)}\bigr)\,. \tag{1}$$

We now compute the rank of the restriction to a single indecomposable $M(a,b)$. By the convention of §5.1, the indecomposable $M(a,b)$ is supported exactly on the cyclic interval $[a \rightsquigarrow b]$; at each vertex in this support the vertex space is one-dimensional, and along each arrow contained in the support the arrow map is the identity, while outside the support the vertex spaces and arrow maps are zero. We claim that

$$\operatorname{rank}\bigl(\Phi_{i \to j}|_{M(a,b)}\bigr) = \begin{cases} 1, & \text{if } [i \rightsquigarrow j] \subseteq [a \rightsquigarrow b], \\ 0, & \text{otherwise.} \end{cases} \tag{2}$$

Indeed, if $[i \rightsquigarrow j] \subseteq [a \rightsquigarrow b]$, then every vertex of the segment $[i \rightsquigarrow j]$ lies in the support of $M(a,b)$, and every arrow along that segment acts as id_k. Hence the composition along the segment is $\mathrm{id}_k : k \to k$, so the rank is 1. Conversely, suppose $[i \rightsquigarrow j] \not\subseteq [a \rightsquigarrow b]$. Then at least one of the following happens:

1. $i \notin [a \rightsquigarrow b]$. Then the source space of $\Phi_{i \to j}|_{M(a,b)}$ is 0, so the map is zero.
2. $j \notin [a \rightsquigarrow b]$. Then the target space is 0, so the map is zero.
3. Both i and j lie in the support, but some intermediate arrow of the segment $[i \rightsquigarrow j]$ does not lie in the support interval $[a \rightsquigarrow b]$. On that arrow the map in $M(a,b)$ is zero, hence the whole composition is zero.

Thus (2) holds. Substituting (2) into (1) gives

$$\operatorname{rank}\bigl(\Phi_{i\to j}(\phi)\bigr) = \sum_{\substack{a,b\\ [i\rightsquigarrow j]\subseteq[a\rightsquigarrow b]}} m_{ab},$$

which is the first formula. It remains to rewrite this in explicit endpoint form. Fix i, j, and let $t := \delta(i, j)$. An interval $[a \rightsquigarrow b]$ contains $[i \rightsquigarrow j]$ if and only if, along the cyclic orientation, one can write

$$a = i - u, \qquad b = j + v$$

for some integers $u, v \geq 0$, and the total length of $[a \rightsquigarrow b]$ is

$$\delta(a, b) = u + t + v \leq L - 1.$$

Equivalently,

$$0 \leq u \leq t, \qquad 0 \leq v \leq L - t - u - 1.$$

Hence

$$r_{ij}(m) = \sum_{u=0}^{t} \sum_{v=0}^{L-t-u-1} m_{\,i-u,\, j+v},$$

with all indices understood modulo $L+1$. This is exactly the stated explicit formula. Finally, since $\operatorname{rank}(\Phi_{i\to j})$ is determined by the multiplicities m_{ab} alone, the array $r(m) = (r_{ij}(m))$ depends only on the orbit data m, not on the chosen representative ϕ. □

Bijection between R_d^{orb} and M_d^+

Proof of lemma 5.17:

Proof. By Definition 5.16,

$$m_{ij}(r) = r_{ij} - r_{i-1,j} - r_{i,j+1} + r_{i-1,j+1},$$

with indices taken modulo $L + 1$, and with the convention that $r_{ab} = 0$ whenever $\delta(a, b) > L$.

Fix i, j with $\delta(i, j) \leq L$. Since $r \in R_d^{\mathrm{orb}}$, the defining inequalities of R_d^{orb} give

$$r_{ij} - r_{i-1,j} - r_{i,j+1} + r_{i-1,j+1} \geq 0.$$

Hence

$$m_{ij}(r) \geq 0 \qquad \text{for all } i, j.$$

It remains to verify the dimension constraint at each vertex p. By the conventions of Section 5.1, the multiplicity array $m = (m_{ab})$ lies in M_d^+ if and only if $m_{ab} \geq 0$ for all a, b, and for every vertex p, the total multiplicity of intervals whose support contains p is equal to d_p, i.e.

$$\sum_{\substack{a,b\\ p\in[a\rightsquigarrow b]}} m_{ab} = d_p.$$

We now check this for $m = G(r)$. Fix p. An interval $[a \rightsquigarrow b]$ contains p if and only if there exist unique integers $u, v \geq 0$ such that

$$a = p - u, \qquad b = p + v, \qquad u + v \leq L,$$

where all indices are taken modulo $L + 1$. Therefore

$$\sum_{\substack{a,b \\ p \in [a \rightsquigarrow b]}} m_{ab}(r) = \sum_{u=0}^{L} \sum_{v=0}^{L-u} m_{p-u,\, p+v}(r).$$

Substituting the definition of $m_{ij}(r)$, this becomes

$$\sum_{u=0}^{L} \sum_{v=0}^{L-u} \big(r_{p-u,p+v} - r_{p-u-1,p+v} - r_{p-u,p+v+1} + r_{p-u-1,p+v+1} \big).$$

For notational convenience, set

$$A_{u,v} := r_{p-u,p+v}.$$

Then the sum is

$$\sum_{u=0}^{L} \sum_{v=0}^{L-u} \big(A_{u,v} - A_{u+1,v} - A_{u,v+1} + A_{u+1,v+1} \big).$$

Rewrite the summand as

$$\big(A_{u,v} - A_{u,v+1} \big) - \big(A_{u+1,v} - A_{u+1,v+1} \big),$$

so that

$$\sum_{u=0}^{L} \sum_{v=0}^{L-u} \big(A_{u,v} - A_{u+1,v} - A_{u,v+1} + A_{u+1,v+1} \big) = \sum_{u=0}^{L} \left[\sum_{v=0}^{L-u} (A_{u,v} - A_{u,v+1}) - \sum_{v=0}^{L-u} (A_{u+1,v} - A_{u+1,v+1}) \right].$$

For each fixed u, the inner sums telescope in v:

$$\sum_{v=0}^{L-u} (A_{u,v} - A_{u,v+1}) = A_{u,0} - A_{u,L-u+1},$$

and similarly

$$\sum_{v=0}^{L-u} (A_{u+1,v} - A_{u+1,v+1}) = A_{u+1,0} - A_{u+1,L-u+1}.$$

Now

$$A_{u,L-u+1} = r_{p-u,\, p+L-u+1} = 0$$

because

$$\delta(p-u,\ p+L-u+1) = L+1,$$

and likewise

$$A_{u+1,L-u+1} = 0.$$

Hence the whole expression reduces to

$$\sum_{u=0}^{L}\bigl(A_{u,0}-A_{u+1,0}\bigr).$$

This now telescopes in u, giving

$$A_{0,0}-A_{L+1,0}.$$

Again,

$$A_{L+1,0}=r_{p-(L+1),\,p}=0$$

by the same convention, since the corresponding oriented segment has length $L+1$. Therefore

$$\sum_{\substack{a,b\\ p\in[a\rightsquigarrow b]}} m_{ab}(r)=A_{0,0}=r_{pp}.$$

Because $r\in R_d\subseteq\mathbb{Z}^{(L+1)\times(L+1)}$, we have $r_{pp}=d_p$. Thus

$$\sum_{\substack{a,b\\ p\in[a\rightsquigarrow b]}} m_{ab}(r)=d_p \qquad \text{for every } p.$$

We have shown that all $m_{ij}(r)$ are nonnegative and that the vertex-dimension constraints are satisfied. Therefore $G(r)\in M_d^+$. □

We prove proposition 5.18:

Proof. We prove the two identities separately.
(i) $G\circ F=\mathrm{id}_{M_d^+}$**.** Let $m\in M_d^+$, and set $r:=F(m)$. By definition, $r=r(m)$ is the rank pattern associated with m. For every pair (i,j) with $\delta(i,j)\le L$, Proposition 5.11 gives

$$m_{ij}=r_{ij}-r_{i-1,j}-r_{i,j+1}+r_{i-1,j+1},$$

with the convention that $r_{ab}=0$ whenever $\delta(a,b)>L$. But the right-hand side is exactly the definition of $m_{ij}(r)$, i.e.

$$m_{ij}(r)=r_{ij}-r_{i-1,j}-r_{i,j+1}+r_{i-1,j+1}.$$

Therefore

$$m_{ij}(r)=m_{ij} \qquad \text{for all } i,j.$$

Equivalently,

$$G(F(m))=m.$$

(ii) $F\circ G=\mathrm{id}_{R_d^{\mathrm{orb}}}$**.** Let $r\in R_d^{\mathrm{orb}}$, and set

$$m:=G(r).$$

By Lemma 5.17, we have $m\in M_d^+$, so $F(m)$ is well-defined. Set

$$r':=F(m)=F(G(r)).$$

We must prove that $r' = r$.

First, for every pair (i,j) with $\delta(i,j) \le L$, Proposition 5.11 applied to the multiplicity array m and its rank pattern $r' = F(m)$ yields

$$m_{ij} = r'_{ij} - r'_{i-1,j} - r'_{i,j+1} + r'_{i-1,j+1}.$$

On the other hand, since $m = G(r)$, Definition 5.16 gives

$$m_{ij} = r_{ij} - r_{i-1,j} - r_{i,j+1} + r_{i-1,j+1}.$$

Hence, for all (i,j) with $\delta(i,j) \le L$,

$$r'_{ij} - r'_{i-1,j} - r'_{i,j+1} + r'_{i-1,j+1} = r_{ij} - r_{i-1,j} - r_{i,j+1} + r_{i-1,j+1}. \tag{*}$$

Second, both r and r' lie in R_d. Indeed, $r \in R_d^{\mathrm{orb}} \subseteq R_d$ by definition, while $r' = F(m) \in R_d^{\mathrm{orb}} \subseteq R_d$ by Lemma 5.15. Therefore

$$r_{ii} = d_i = r'_{ii} \qquad \text{for all } i. \tag{**}$$

We now prove $r'_{ij} = r_{ij}$ for all i,j with $\delta(i,j) \le L$, by induction on $\ell := \delta(i,j)$.

If $\ell = 0$, then $j = i$, and the claim is exactly $(**)$:

$$r'_{ii} = d_i = r_{ii}.$$

Assume now that $\ell \ge 1$, and that $r'_{ab} = r_{ab}$ for all pairs (a,b) with $\delta(a,b) < \ell$. Fix i,j with $\delta(i,j) = \ell$. From $(*)$ we have

$$r'_{ij} - r'_{i-1,j} - r'_{i,j+1} + r'_{i-1,j+1} = r_{ij} - r_{i-1,j} - r_{i,j+1} + r_{i-1,j+1}.$$

Rearranging,

$$r'_{ij} - r_{ij} = \big(r'_{i-1,j} - r_{i-1,j}\big) + \big(r'_{i,j+1} - r_{i,j+1}\big) - \big(r'_{i-1,j+1} - r_{i-1,j+1}\big). \tag{***}$$

Now observe that

$$\delta(i-1,j) = \ell - 1, \qquad \delta(i,j+1) = \ell - 1, \qquad \delta(i-1,j+1) = \ell - 2,$$

where the last term is interpreted only when $\ell \ge 2$; if $\ell = 1$, then $\delta(i-1,j+1) = L+1$, and by convention both $r_{i-1,j+1}$ and $r'_{i-1,j+1}$ are 0. In all cases, every term on the right-hand side of $(***)$ involves a pair of strictly smaller segment length than ℓ. By the induction hypothesis, each of those differences is zero. Hence

$$r'_{ij} - r_{ij} = 0.$$

Therefore $r'_{ij} = r_{ij}$.

By induction, $r' = r$ for all pairs (i,j) with $\delta(i,j) \le L$. Thus

$$F(G(r)) = r.$$

Combining (i) and (ii), we conclude that F and G are mutually inverse bijections. $\square$

Stratification of cyclic conditions with target map by rank patterns

Proof of Proposition 5.20. Fix $r \in R_d^{\mathrm{orb}}$. By Proposition 5.18, the maps

$$F : M_d^+ \to R_d^{\mathrm{orb}}, \qquad G : R_d^{\mathrm{orb}} \to M_d^+$$

are mutually inverse. Therefore there exists a unique multiplicity array

$$m := G(r) \in M_d^+$$

such that

$$F(m) = r.$$

Let $\mathcal{O}_m \subset \Gamma_{d,L+1}$ denote the corresponding G_d-orbit. Since $F(m) = r$, the cyclic rank pattern of every element of $\mathcal{O}_m$ is r. Hence $\mathcal{O}_m = \mathcal{O}_r$.

By Lemma 5.3, for any G_d-orbit $\mathcal{O} \subset \Gamma_{d,L+1}$, the intersection

$$\mathcal{O} \cap \Gamma_{d,\sigma}$$

is either empty or a single G_σ-orbit. Applying this to $\mathcal{O} = \mathcal{O}_r$, we obtain that

$$\mathcal{O}_{r,\sigma} = \mathcal{O}_r \cap \Gamma_{d,\sigma}$$

is either empty or a single G_σ-orbit.

It remains to characterize when $\mathcal{O}_{r,\sigma}$ is nonempty. Since $\mathcal{O}_r = \mathcal{O}_m$, Proposition 5.4 gives

$$\mathcal{O}_{r,\sigma} \neq \varnothing \quad \Longleftrightarrow \quad \mathcal{O}_m \cap \Gamma_{d,\sigma} \neq \varnothing \quad \Longleftrightarrow \quad \mathrm{rank}(m) = \mathrm{rank}(\sigma),$$

where $\mathrm{rank}(m)$ denotes the rank label of the orbit $\mathcal{O}_m$.

By Proposition 5.7, this rank label is given by the activity formula

$$\mathrm{rank}(m) = \sum_{\substack{i,j \\ \delta(i,j)\geq 1}} \epsilon(i,j)\, m_{ij}.$$

Since $m = G(r)$, we have

$$m_{ij} = m_{ij}(r) = r_{ij} - r_{i-1,j} - r_{i,j+1} + r_{i-1,j+1},$$

so

$$\mathrm{rank}(m) = \sum_{\substack{i,j \\ \delta(i,j)\geq 1}} \epsilon(i,j)\, m_{ij}(r) = \sum_{\substack{i,j \\ \delta(i,j)\geq 1}} \epsilon(i,j)\, \big(r_{ij} - r_{i-1,j} - r_{i,j+1} + r_{i-1,j+1}\big).$$

This is exactly the definition of $\mathrm{rank}(r)$. Therefore

$$\mathcal{O}_{r,\sigma} \neq \varnothing \quad \Longleftrightarrow \quad \mathrm{rank}(r) = \mathrm{rank}(\sigma).$$

This proves the proposition. $\square$

Proof of Theorem 5.21. Since $\Gamma_{d,L+1}$ is the disjoint union of its G_d-orbits and Proposition 5.18 identifies those orbits with cyclic rank patterns $r \in R_d^{\mathrm{orb}}$, we have the disjoint decomposition

$$\Gamma_{d,L+1} = \bigsqcup_{r \in R_d^{\mathrm{orb}}} \mathcal{O}_r.$$

Intersecting both sides with $\Gamma_{d,\sigma}$ yields

$$\Gamma_{d,\sigma} = \Gamma_{d,\sigma} \cap \Gamma_{d,L+1} = \bigsqcup_{r \in R_d^{\mathrm{orb}}} \big(\mathcal{O}_r \cap \Gamma_{d,\sigma}\big) = \bigsqcup_{r \in R_d^{\mathrm{orb}}} \mathcal{O}_{r,\sigma}.$$

This union is disjoint because the G_d-orbits $\mathcal{O}_r$ are pairwise disjoint.

Now apply Proposition 5.20. For a given $r \in R_d^{\mathrm{orb}}$, the term $\mathcal{O}_{r,\sigma}$ is empty unless

$$\mathrm{rank}(r) = \mathrm{rank}(\sigma).$$

Hence all empty terms may be removed, and we obtain

$$\Gamma_{d,\sigma} = \bigsqcup_{\substack{r \in R_d^{\mathrm{orb}} \\ \mathrm{rank}(r) = \mathrm{rank}(\sigma)}} \mathcal{O}_{r,\sigma}.$$

Again by Proposition 5.20, each nonempty $\mathcal{O}_{r,\sigma}$ is a single G_σ-orbit. Therefore $\Gamma_{d,\sigma}$ is stratified by G_σ-orbits indexed by those cyclic rank patterns $r \in R_d^{\mathrm{orb}}$ satisfying $\mathrm{rank}(r) = \mathrm{rank}(\sigma)$. $\square$